



CAPSTONE PROJECT FINAL REPORT PGP-DSBA

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INTRODUCTION OF THE BUSINESS PROBLEM:

PROBLEM STATEMENT:

An E Commerce company provider is facing a lot of competition in the current market and it has become a challenge to retain the existing customers in the current situation. Hence, the company wants to develop a model through which they can do churn prediction of the accounts and provide segmented offers to the potential churners. In this company, account churn is a major thing because 1 account can have multiple customers. hence by losing one account the company might be losing more than one customer. You have been assigned to develop a churn prediction model for this company and provide business recommendations on the campaign.

NEED OF THE PROJECT:

This study is vital for the client to gain a deeper understanding of the customer lifecycle, uncover churn patterns, and pinpoint the key drivers behind customer attrition. With these insights, the company can make data-driven decisions to proactively retain similar customer profiles in the future.

Given that each account may serve multiple customers, losing a single account could result in the loss of several users—making churn prevention even more critical.

By identifying the factors that contribute to churn early, the company can design strategic retention initiatives before the customer disengages completely.

UNDERSTANDING BUSINESS OPPORTUNITY:

This e-commerce company operates on a model where users create individual profiles to make purchases online. The platform is widely appreciated for its round-the-clock accessibility and global reach—customers can shop at any time and have products delivered from virtually anywhere.

Notably, a single account may be used by multiple individuals for separate transactions. For instance, if Person A has an account, Person B may use it to make purchases, highlighting a flexible and shared user experience. The core objective of the business is to simplify the shopping journey, allowing customers to complete purchases quickly and conveniently, without spending long hours as they might in a physical store.

Given this structure, retaining every customer who enrolls on the platform becomes crucial. Each active account represents ongoing potential revenue, as customers are likely to continue engaging and transacting over time.

DATA INFORMATION:

```
RangeIndex: 11260 entries, 0 to 11259
Data columns (total 19 columns):
#   Column                Non-Null Count  Dtype
---  -
0   AccountID              11260 non-null  int64
1   Churn                  11260 non-null  int64
2   Tenure                 11158 non-null  object
3   City_Tier              11148 non-null  float64
4   CC_Contacted_LY        11158 non-null  float64
5   Payment                11151 non-null  object
6   Gender                 11152 non-null  object
7   Service_Score          11162 non-null  float64
8   Account_user_count      11148 non-null  object
9   account_segment        11163 non-null  object
10  CC_Agent_Score          11144 non-null  float64
11  Marital_Status          11048 non-null  object
12  rev_per_month           11158 non-null  object
13  Complain_ly            10903 non-null  float64
14  rev_growth_yoy         11260 non-null  object
15  coupon_used_for_payment 11260 non-null  object
16  Day_Since_CC_connect    10903 non-null  object
17  cashback                10789 non-null  object
18  Login_device           11039 non-null  object
dtypes: float64(5), int64(2), object(12)
memory usage: 1.6+ MB
```

Fig 1: Data information

Observation:

- Account ID can be excluded from the analysis.
- Out of the 19 columns, only four columns have complete data—the rest contain missing values.
- The dataset includes 5 float columns, 2 integer columns, and 12 object (string) columns.
- Among the 12 object columns, the following features appear to be misclassified and should ideally be numeric:
 - Tenure
 - Account_user_count
 - rev_per_month
 - rev_growth_yoy
 - coupon_used_for_payment
 - Day_Since_CC_connect
 - cashback

These will need appropriate type conversion and validation.

DATA INTEGRITY:

- City_Tier should be converted as Object for categorical representation.
- Complains_ly has only binary value hence we can change it as object for categorical representation.
- The Tenure column contains the special character # which has to be converted as nan value.
- The Gender column includes mixed representations such 'F', and 'M' this should be converted to Male and Female.
- The Account_user_count field contains the @ symbol which has to be converted as nan value.
- rev_per_month has values with a + symbol that need to be cleaned for numeric conversion.
- rev_growth_yoy includes entries with the \$ symbol which has to be converted as nan value.
- The coupon_used_for_payment field contains various symbols like \$, #, and * which has to be converted as nan value.
- Day_Since_CC_connect also includes values with the \$ symbol which has to be converted as nan value.
- The Login_device column contains entries like '&&&&' which has to be converted as nan value.
- account_segment has 'Regular +, Super + which has been changed to Regular plus and Super Plus.
- Service_Score - Rating of 0 is not typically valid in most scoring systems, these entries likely represent missing or invalid responses and should be treated as non-values (Nan) for appropriate handling in subsequent analysis

Apart from this most of the columns have nan values, which can be treated while performing missing value treatment.

STATISTICAL SUMMARY WITH VARIABLE INFO:

Numeric Data:

	count	mean	std	min	25%	50%	75%	max
AccountID	11260.00	25630.00	3251.00	20000.00	22815.00	25630.00	28444.00	31259.00
Tenure	11042.00	11.00	13.00	0.00	2.00	9.00	16.00	99.00
CC_Contacted_LY	11158.00	18.00	9.00	4.00	11.00	16.00	23.00	132.00
Service_Score	11154.00	3.00	1.00	1.00	2.00	3.00	3.00	5.00
Account_user_count	10816.00	4.00	1.00	1.00	3.00	4.00	4.00	6.00
CC_Agent_Score	11144.00	3.00	1.00	1.00	2.00	3.00	4.00	5.00
rev_per_month	10469.00	6.00	12.00	1.00	3.00	5.00	7.00	140.00
rev_growth_yoy	11257.00	16.00	4.00	4.00	13.00	15.00	19.00	28.00
coupon_used_for_payment	11257.00	2.00	2.00	0.00	1.00	1.00	2.00	16.00
Day_Since_CC_connect	10902.00	5.00	4.00	0.00	2.00	3.00	8.00	47.00
cashback	10787.00	196.00	179.00	0.00	147.00	165.00	200.00	1997.00

Table 1: Numeric column Statistical Summary

Observation:**Tenure:**

Based on the available data, the average tenure is 11 months, with a minimum of 0 months, indicating that some users may be churning before completing even one full month. The maximum tenure observed is 99 months. Since the mean is greater than the median, the distribution is right-skewed, and the data shows a standard deviation of 12 months.

CC_Contacted_LY:

On average, all customers associated with an account contacted customer care 18 times over the past 12 months. The minimum number of contacts was 4, and the maximum reached 132. Notably, the large gap between the 75th percentile and the maximum value indicates the presence of high outliers. Since the mean is less than the median, the distribution is left-skewed, suggesting that while most users cluster around higher values, a significant portion made relatively few contacts.

Service_Score:

The Service_score ranges from 1 to 5, with a mean of 3. Approximately 75% of customers rated the service as 3 or below, indicating that higher satisfaction levels are relatively rare.

Account_user_count:

On average, 4 customers are tagged to a single account, with the minimum being 1 and the maximum reaching 6. Approximately 75% of accounts have 4 or fewer customers associated with them. This suggests a fairly compact distribution, with the majority of accounts hosting small customer groups.

CC_Agent_Score:

The CC_Agent_Score ranges from 1 to 5, with a mean of 3, indicating a moderate average perception of agent performance. Notably, in 75% of the cases, the score is 4 or below, suggesting that higher ratings are less frequent and that the majority of customers rate their customer care experience as average or slightly above average.

rev_per_month:

Assuming the revenue is reported in USD, the rev_per_month column shows a mean of \$6, with values ranging from a minimum of \$1 to a maximum of \$140. Since the mean is greater than the median, the distribution is right-skewed, suggesting that a small number of high-revenue observations are pulling the average upward. The standard deviation is \$12, indicating substantial variability in customer revenue across accounts.

rev_growth_yoy:

The rev_growth_yoy column shows a mean of \$16, with values ranging from a minimum of \$4 to a maximum of \$28. Since the mean is greater than the median, the distribution is right-skewed, suggesting that a small number of high-revenue observations are pulling the average upward. The standard deviation is \$4.

coupon_used_for_payment:

Over the past 12 months, customers have used coupons for payments an average of 2 times, with usage ranging from a minimum of 0 to a maximum of 16 times. Despite the maximum being relatively high, approximately 75% of customers have used coupons 2 times or fewer, indicating that frequent coupon usage.

Day_Since_CC_connect:

The number of days since no customers contacted customer care ranges from a minimum of 0 days to a maximum of 47 days. The average (mean) is 5 days, while 75% of the time, the count is 8 days or fewer.

Cashback:

Over the last 12 months, the cashback generated per account ranges from a minimum of 0 to a maximum of \$1997. While the mean cashback is notably greater than the median, indicating a right-skewed distribution, approximately 75% of accounts received 200 or less in cashback. The data shows a standard deviation of \$179.

Object Data:

	count	unique	top	freq
Churn	11260	2	0	9364
City_Tier	11148.00	3.00	1.00	7263.00
Payment	11151	5	Debit Card	4587
Gender	11152	2	Male	6704
account_segment	11163	5	Regular Plus	4124
Marital_Status	11048	3	Married	5860
Complain_ly	10903.00	2.00	0.00	7792.00
Login_device	10500	2	Mobile	7482

Table 2: Object column Statistical Summary

Observation

Churn:

The target column is binary, with values 0 and 1. A value of 1 indicates customers who have churned, whereas 0 represents those who have retained. The dataset is imbalanced, with a majority of records labeled as 0, suggesting that most customers in the dataset did not churn.

City_Tier:

The City_Tier column consists of three distinct tiers: 1, 2, and 3. Among these, Tier 3 accounts for the largest share of customers.

Payment:

The dataset includes five distinct payment modes, among which Debit Card emerges as the most widely preferred option by customers

Gender:

The dataset shows a higher number of male customers compared to female customers.

account_segment:

The dataset includes five distinct account segments, among which 'Regular Plus' is the most preferred by customers.

Marital_Status:

The dataset includes customers across three marital statuses, with Married individuals forming the largest group.

Complain_ly:

This column contains binary values, where 1 indicates that a complaint was raised by the customer in the past 12 months, and 0 indicates no complaint. The data shows that the majority of customers have not raised any complaints.

Login_device:

The dataset features two types of login devices, with Mobile being the most commonly used among customers.

DUPLICATE DETECTION:

Since each row represents unique Account number this data set has no duplicate data.

EXPLORATORY DATA ANALYSIS:

UNIVARIATE ANALYSIS:

OBJECT COLUMNS:

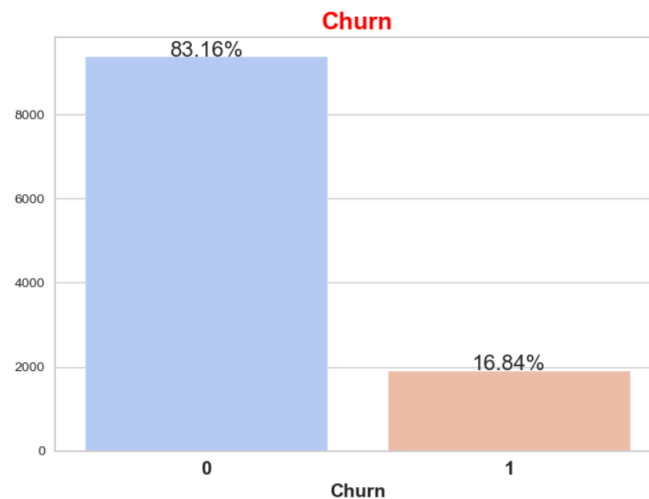


Fig 2: Churn Column

- 83.16% of customers do not churn, while only 16.84% do. This indicates a significant imbalance in the dataset.

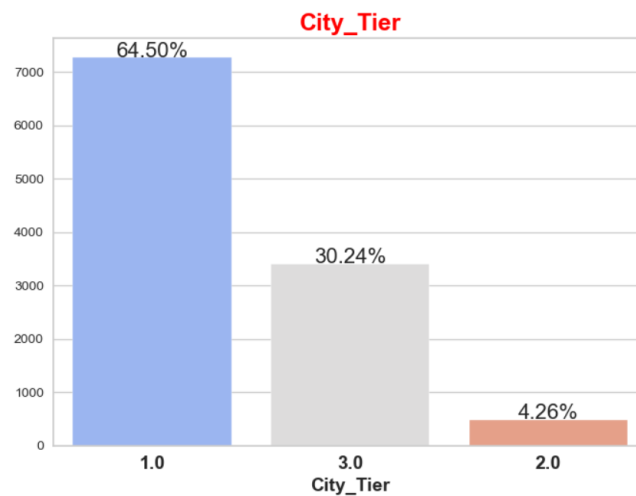


Fig 3: City_Tier Column

- The majority of customers (64.50%) are concentrated in City Tier 1, followed by 30.24% in City Tier 3.
- City Tier 2 accounts for only a small share, with just 4.26% of the customer base.

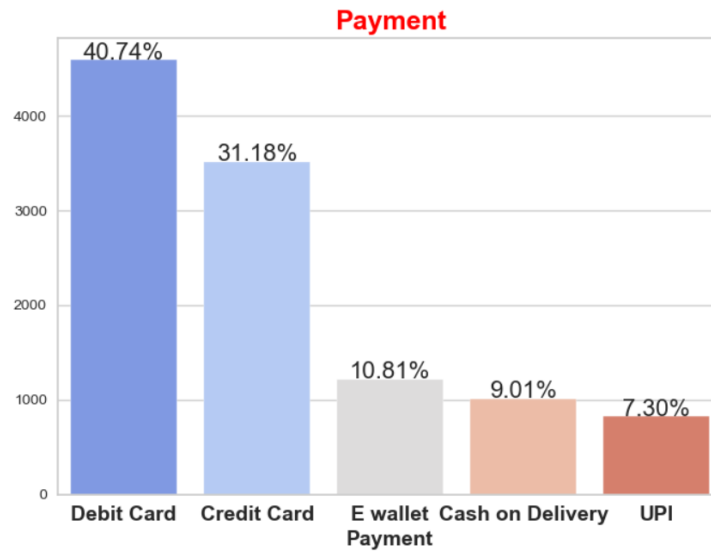


Fig 4: Payment Column

- Debit cards are the most preferred payment mode among customers, with 40.74% opting for them. Credit cards follow at 31.18%.
- UPI lags significantly behind, chosen by only 7.30% of customers.

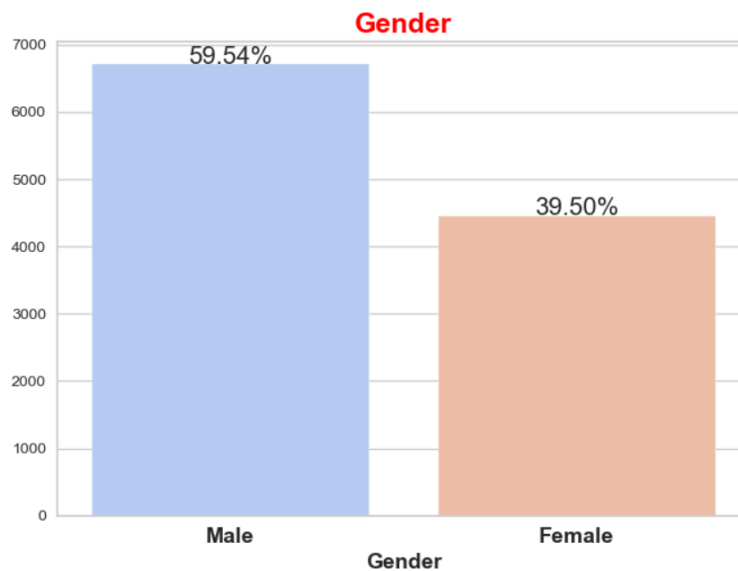


Fig 5: Gender Column

- Male customers constitute the majority at 59.54%, while female customers represent 39.50% of the customer base.

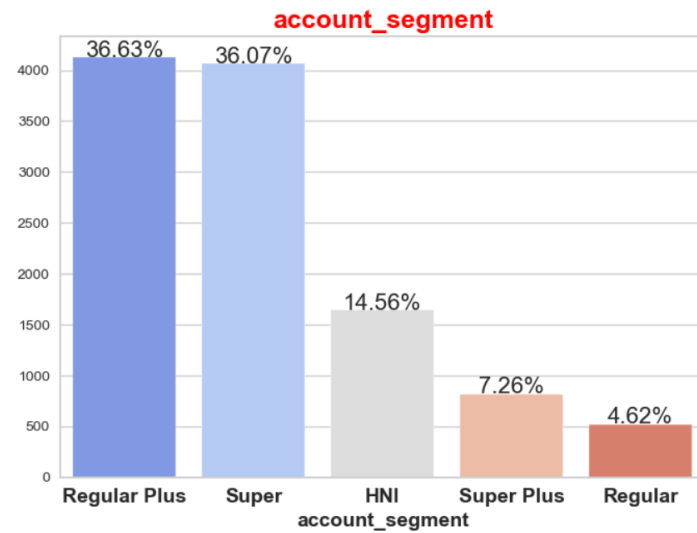


Fig 6: Account_segment Column

- Regular Plus is the most preferred account segment, chosen by 36.63% of customers, closely followed by the Super segment at 36.07%.
- Together, they represent 72.7% of the customer base.
- The remaining customers fall into the HNI, Super Plus, and Regular segments.

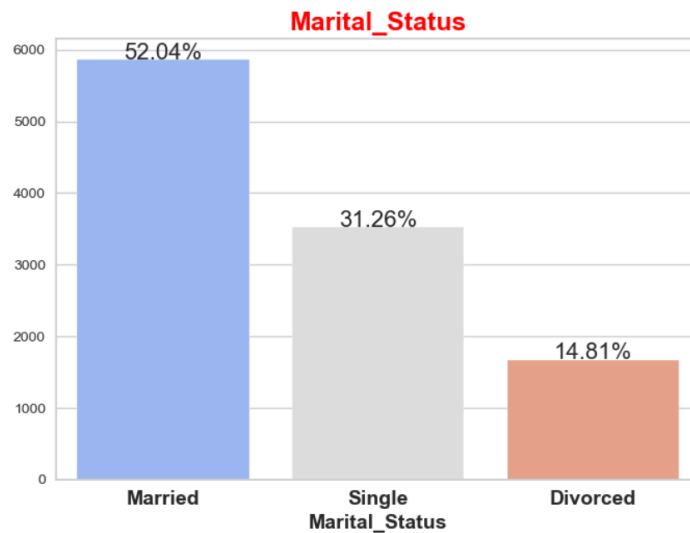


Fig 7: Marital_status Column

- A majority of customers, 52.04%, are married.
- Single individuals make up 31.26% of the base, while the remaining customers are divorced.

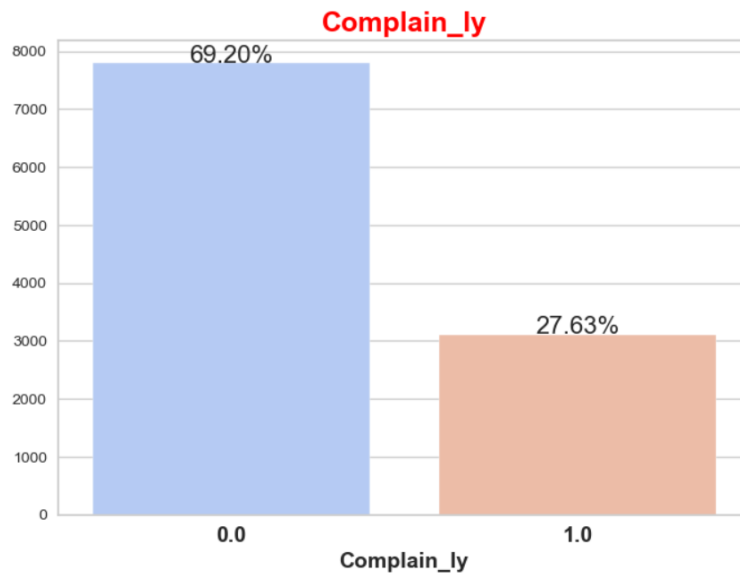


Fig 8: Complain_Iy Column

- The majority of customers (69.20%) have not raised any complaints, while only 27.63% have lodged a complaint within the past 12 months.

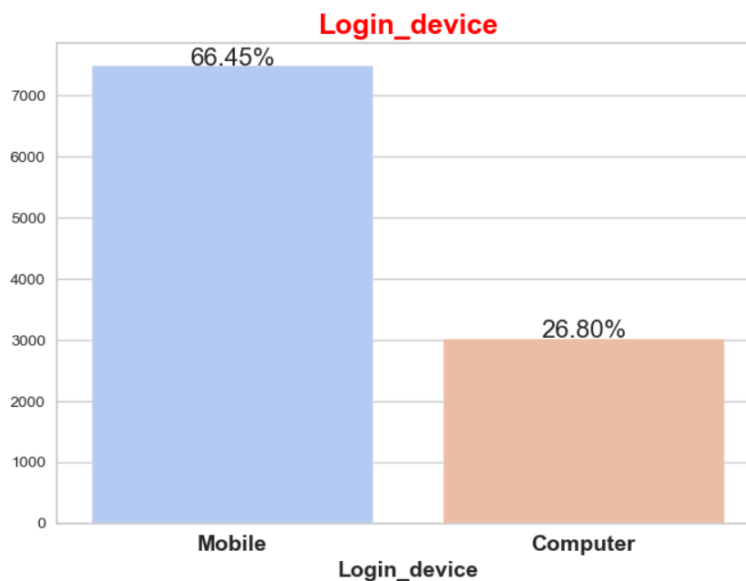


Fig 9: Login_Device Column

- A majority of customers (66.45%) use mobile devices to log in, while the remaining users access their accounts via computer.

CATEGORICAL COLUMNS VS CHURN COLUMN:

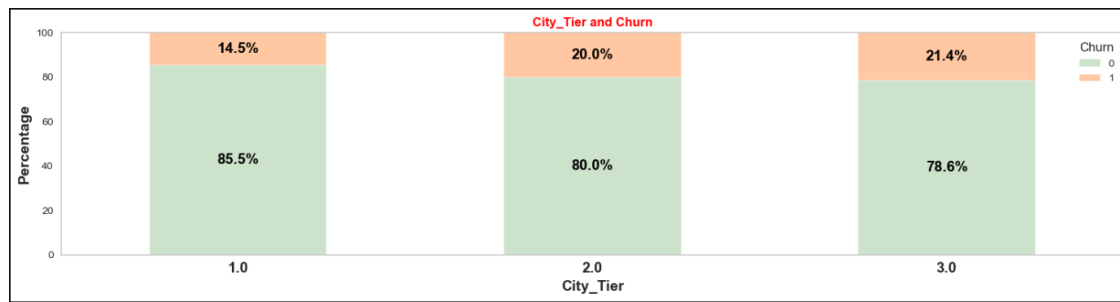


Fig 10: City_Tier vs Churn

Observation:

- Customer churn is highest in Tier 3 cities at 21.4%, followed by Tier 2 cities at 20%. Tier 1 cities have the lowest churn rate among the three, at 14.5%.

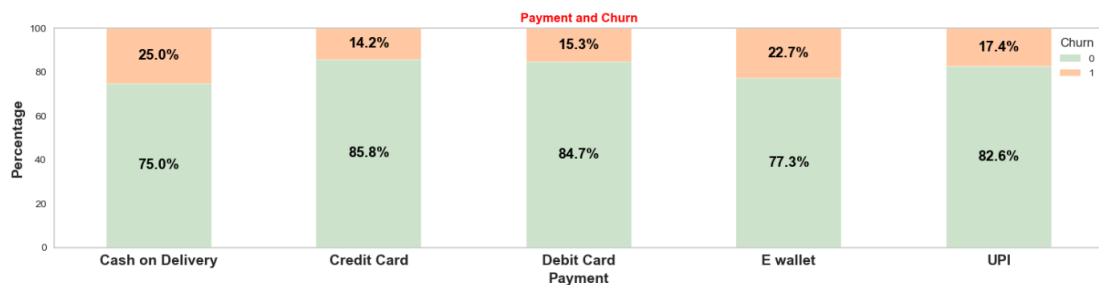


Fig 11: Payment vs Churn

Observation:

- Customers opting for Cash on Delivery exhibit the highest churn rate at 25%, followed by those using E-wallets at 22.7% and UPI at 17.4%. In contrast, churn rates among Credit Card and Debit Card users are comparatively lower.

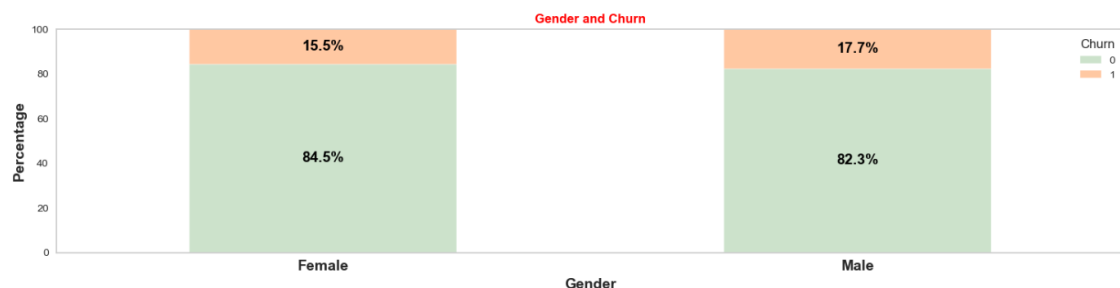


Fig 12: Gender vs Churn

- Churn rates among male customers stand at 17.7%, while female customers show a slightly lower rate of 15.5%, indicating minimal variation in behavior across genders.

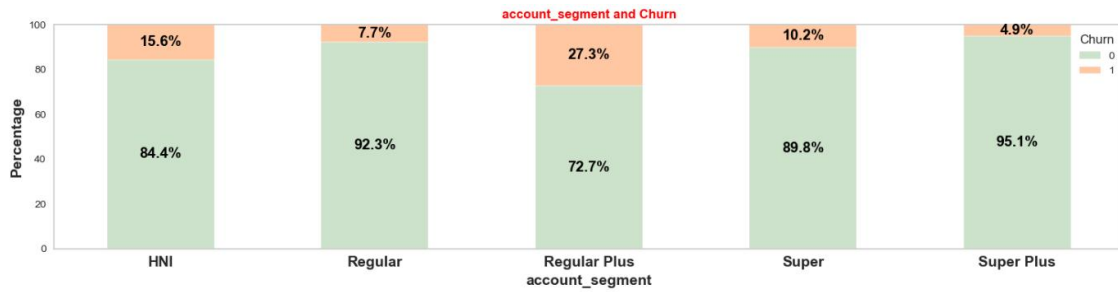


Fig 13: account_segment vs Churn

Observation:

- Despite being the most preferred, the RegularPlus segment experiences the highest customer churn rate at 27.3%.
- It's followed by the HNI and Super segments. In contrast, the Regular and Super Plus segments exhibit significantly lower churn rates.

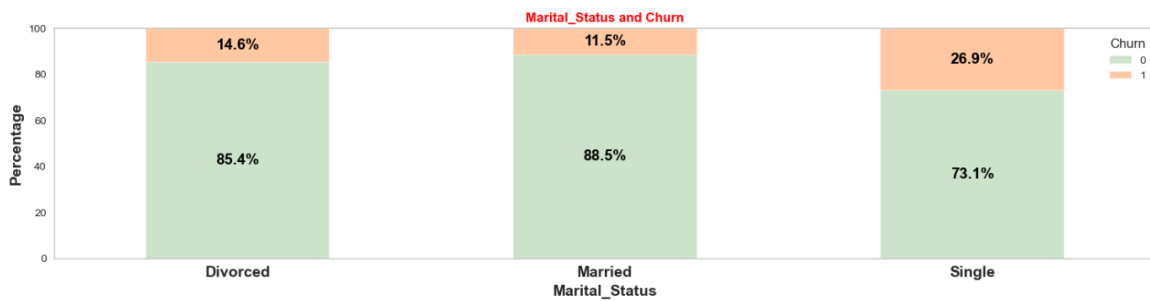


Fig 14: Marital_Status vs Churn

Observation:

- Single customers exhibit the highest churn rate at 26.9%, followed by divorced customers at 14.6%. Despite accounting for 53% of the overall customer base, married customers show a notably lower churn rate of just 11.5%.
- This pattern suggests that marital status may be an influencing factor in customer retention.

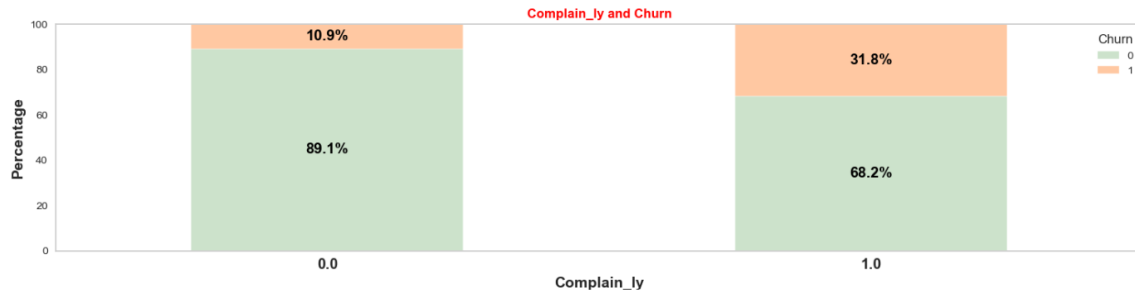


Fig 15: Complain_ly vs Churn

Observations:

- Customers who have filed complaints within the past 12 months show the highest churn rate at 31.8%, indicating a strong correlation between dissatisfaction and customer attrition.

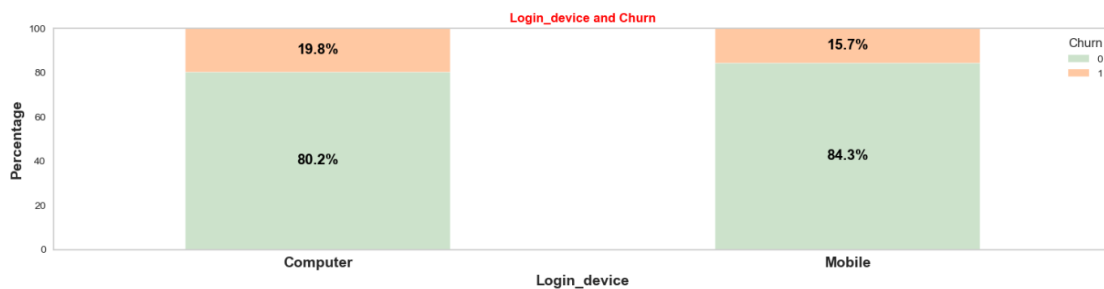


Fig 16: Login_Device vs Churn

Observations:

- Customers accessing the platform via computers exhibit the highest churn rate at 19.8%, compared to a lower rate of 15.7% among mobile users.

NUMERIC COLUMNS VS TARGET COLUMNS:

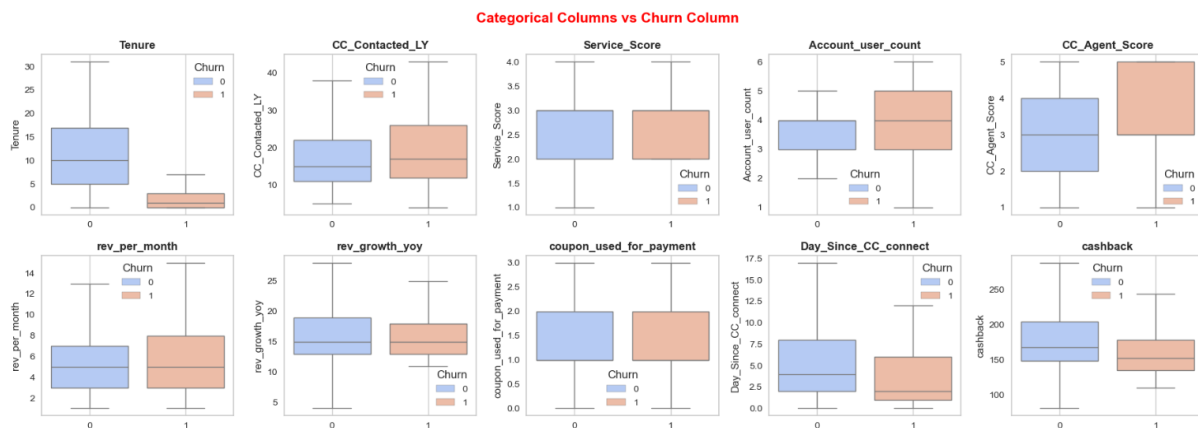


Fig 17: Numeric columns vs Churn

Observation:

- Customers with a tenure of 5 months or less are more likely to churn.
- The 75th percentile of customer care interactions last year is noticeably higher among churned customers than those who retained their accounts.
- Accounts with either fewer than 2 users or more than 4 are also more prone to churn.
- Additionally, the average number of days since the last customer care interaction is higher for non-churning customers, indicating that frequent engagement with customer care is associated with higher churn risk.
- Lastly, churned customers tend to receive less cashback compared to those who remain.

MULTI VARIATE ANALYSIS:

PAYMENT, CITY_TIER AND CASHBACK BEHAVIOR ANALYSIS:

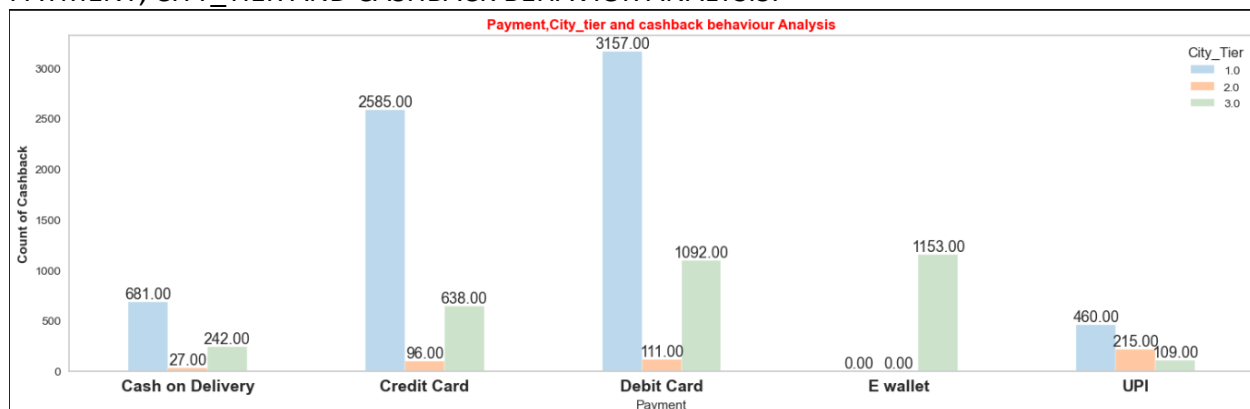


Fig 18: Payment, City_tier and cashback behavior Analysis

Observation:

- In City Tier 1, customers who preferred Debit Cards received cashback frequently, whereas those opting for E-Wallets received none.
- In City Tier 2, frequent cashback was observed among UPI users, with E-Wallet users again receiving none.
- Interestingly, in City Tier 3, the pattern shifts—E-Wallet users received the most cashback, followed by those using Debit Cards.

Additional EDA details are provided in **Appendix A**, beginning on page 26.

DATA PREPROCESSING

REMOVAL OF UNWANTED VARIABLES:

ACCOUNTID – This column is removed already, as it doesn't add any value to our analysis.

rev_per_month – We removed rev_per_month because it had missing data and wasn't needed—
rev_growth_yoy already captures revenue trends for churn analysis.

DATA PREPARATION FOR MODEL

The Proportion and Churning and non-churning customers are imbalanced in the dataset. This is addressed with oversampling technique called **SMOTE**.

0 (non-churn)– 6253

1 (churn)- 6253

VARIABLE TRANSFORMATION:

Tenure, CC_Contacted_LY, coupon_used_for_payment, rev_growth_yoy, Day_Since_CC_connect, Cashback are transformed with log1p.

Heatmap

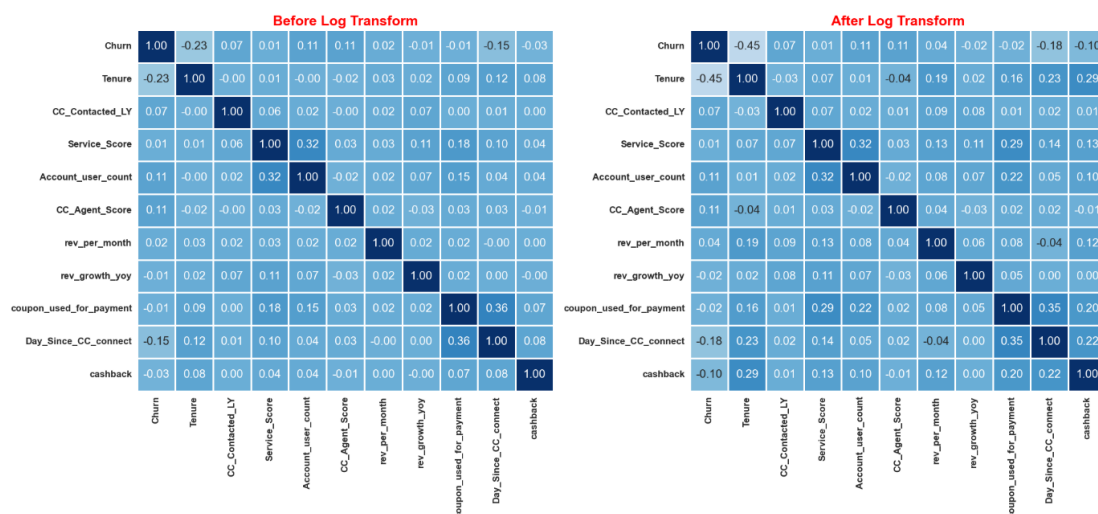


Fig 19: Heat map

Observation:

- Log transformation helped strengthen correlations across multiple features, revealing deeper relationships in the data.
- Among all variables, Tenure shows the strongest (moderate) negative correlation with Churn.
- This means: as a customer's tenure increases, their likelihood of churning decreases, underscoring tenure as a key retention indicator.

MISSING VALUE TREATMENT OF OBJECT COLUMNS:

- All object-type columns were imputed with 'Others' as a placeholder value, except for Gender and Complain_ly, which are binary. For these two, null entries were removed instead of imputed.
- Then dummy variables are created with X data.
- Columns containing the string 'Others' have been removed to prevent imputed values from introducing bias into the model.

This approach retains most rows while avoiding imputation bias from using the mode. By tagging missing values as 'Others' and dropping them later, we preserve row count without skewing the data.

SPLITTING DATA:

The independent variable(X) and target variable (y) have now been split into training, validation, and test datasets.

96% of the overall Data is used for Modeling.

MISSING VALUE TREATMENT FOR NUMERIC COLUMNS:

To handle missing values in the numeric columns, a KNN imputer was applied. It was fit and transformed on the training dataset (X_train), and then used to transform the validation (X_val) and test datasets (X_test).

	Tenure	CC_Contacted_LY	Service_Score	Account_user_count	CC_Agent_Score	rev_per_month	rev_growth_yoy	coupon_used_for_payment	Day_Since_CC_connect
4915	3.00	2.94	3.00	4.00	1.00	1.79	2.71	0.69	1.10
8731	3.47	2.83	3.00	4.00	5.00	2.20	2.64	0.69	2.48
6489	0.69	3.64	2.00	4.00	5.00	1.61	2.64	0.69	1.61
7618	2.77	2.71	3.00	1.00	1.00	1.61	2.71	0.69	1.39
6831	0.00	2.48	2.00	4.00	4.00	2.56	2.64	1.10	1.10

Table 3: Overview after missing value treatment.

SCALING DATASET:

StandardScaler was used to scale the training, validation, and test datasets. It was fit and transformed on the training data, and then applied to the validation and test sets for consistent scaling.

	Tenure	CC_Contacted_LY	Service_Score	Account_user_count	CC_Agent_Score	rev_per_month	rev_growth_yoy	coupon_used_for_payment	Day_Since_CC_connect
4915	0.93	0.26	0.14	0.31	-1.51	0.08	-0.55	-0.27	-0.54
8731	1.36	0.01	0.14	0.31	1.40	0.82	-0.87	-0.27	1.36
6489	-1.19	1.79	-1.26	0.31	1.40	-0.25	-0.87	-0.27	0.16
7618	0.72	-0.27	0.14	-2.65	-1.51	-0.25	-0.55	-0.27	-0.15
6831	-1.83	-0.76	-1.26	0.31	0.67	1.48	-0.87	0.44	-0.54

Table 4: Scaling data.

CLUSTERING USING K – MEANS ALGORITHM WITH K = 2:

K_means_segments	0	1
Tenure	14.000000	9.000000
CC_Contacted_LY	19.000000	18.000000
Service_Score	4.000000	3.000000
Account_user_count	5.000000	4.000000
CC_Agent_Score	4.000000	4.000000
rev_growth_yoy	17.000000	16.000000
coupon_used_for_payment	3.000000	1.000000
Day_Since_CC_connect	6.000000	4.000000
cashback	200.000000	155.000000
count_in_each_segment	5823.000000	4972.000000

Table 5: Clusters.

Cluster 0 Overview — High-Value, Loyal Customers (5,823 individuals)

This segment consists of customers who consistently deliver long-term value to the company and show strong signs of retention. On average, their tenure spans ****14 months****, indicating a deep and ongoing engagement.

Cluster 1 – Short-Term, Revenue-Driven Customers (4,972 total)

This segment contributes meaningfully to revenue but tends to have shorter engagement and a higher churn rate. On average, customers in this group stay for 9 months.

MODEL BUILDING:

Our churn prediction models are trained on two distinct datasets:

Actual Data: The original dataset with natural class imbalance

Oversampled Data: A balanced version created using SMOTE to enhance the model's ability to detect churn cases

MODEL CAN MAKE WRONG PREDICTIONS AS:

False Positive: The model predicts the customer will churn, but in reality, customer does not.

False Negative: The model predicts the customer will not churn, but in reality, customer does.

We aim to maximize **recall**, ensuring the model effectively identifies customers at risk of churn. This approach prioritizes capturing true churn cases, even if it results in a few false alarms, enabling timely and targeted retention efforts

PARAMETRIC MODELS:

- **Logistic Regression:** Assumes the data has a linear pattern, features aren't too closely related (no multicollinearity), and sometimes expects errors to follow a normal distribution.
- **Naive Bayes:** Assumes all features are independent from each other once the class label is known.
- **Linear Discriminant Analysis (LDA):** Assumes that each class generates data based on a Gaussian distribution with a shared covariance matrix. It seeks linear combinations of features that best separate the classes by maximizing the ratio of between-class variance to within-class variance.

NON-PARAMETRIC MODELS.

- **KNNClassifier:** Classifies based on the majority label among nearby data points.
- **DecisionTree:** Splits data using feature thresholds for pure class outcomes.
- **BaggingClassifier:** Combines predictions from multiple best with unstable base learners like trees.
- **RandomForest:** A type of bagging with randomized feature selection at each split.

- **AdaBoostClassifier:** Sequentially focuses on misclassified samples, adjusting weights to improve learning.
- **GBMClassifier:** Builds trees sequentially, correcting errors at each stage.
- **XGBClassifier:** An optimized version of GBM with regularization, handling sparsity, and fast computation.

MODELS & ITS PERFORMANCE:

Model details are presented in the **Appendix B** section beginning on page 34.

Actual Data											
Models	Train					Validation					Performance
	Accuracy	Recall	Precision	F1-Score	AUC Score	Accuracy	Recall	Precision	F1-Score	AUC Score	
Logistic Regression	89.59%	55.49%	76.14%	64.20%	76.00%	90.44%	56.47%	80.93%	66.53%	77.00%	Underfitting
Naïve Bayes	80.98%	70.83%	45.76%	55.60%	77.00%	80.99%	73.38%	45.95%	56.51%	78.00%	Underfitting
LDA	89.72%	60.16%	73.88%	66.32%	78.00%	90.19%	59.35%	77.10%	67.07%	78.00%	Underfitting
KNN	98.06%	91.70%	96.59%	94.08%	96.00%	95.22%	77.70%	92.70%	84.54%	88.00%	Overfitting
Decision tree	100.00%	100.00%	100.00%	100.00%	100.00%	96.49%	88.13%	90.74%	89.42%	93.00%	Overfitting
Bagging	99.79%	98.81%	99.92%	99.36%	99.00%	97.58%	89.57%	95.77%	92.57%	94.00%	Overfitting
Random Forest	100.00%	100.00%	100.00%	100.00%	100.00%	97.76%	88.49%	98.01%	93.01%	94.00%	Overfitting
Ada Boost	89.07%	56.36%	72.53%	63.43%	76.00%	88.92%	56.83%	71.49%	63.33%	76.00%	Underfitting
Gradient Boost	91.84%	62.85%	84.66%	72.14%	80.00%	91.28%	60.43%	83.17%	70.00%	79.00%	Underfitting
XG Boost	99.99%	99.92%	100.00%	99.96%	100.00%	97.46%	89.93%	94.70%	92.25%	94.00%	Overfitting
Oversampled data											
Models	Train					Validation					Performance
	Accuracy	Recall	Precision	F1-Score	AUC Score	Accuracy	Recall	Precision	F1-Score	AUC Score	
Logistic Regression	80.45%	80.12%	80.65%	80.38%	80.00%	80.57%	81.29%	45.66%	58.47%	81.00%	Balanced
Naïve Bayes	74.81%	84.56%	70.76%	77.05%	75.00%	67.19%	85.25%	32.11%	46.65%	74.00%	Balanced
LDA	80.47%	79.98%	80.78%	80.38%	80.00%	80.39%	81.29%	45.38%	58.25%	81.00%	Balanced
KNN	98.35%	99.98%	96.81%	98.37%	98.00%	93.83%	98.56%	73.66%	84.31%	96.00%	Overfitting
Decision tree	100.00%	100.00%	100.00%	100.00%	100.00%	93.95%	86.69%	79.28%	82.82%	91.00%	Overfitting
Bagging	99.94%	99.90%	99.97%	99.94%	100.00%	95.70%	87.41%	87.10%	87.25%	92.00%	Overfitting
Random Forest	100.00%	100.00%	100.00%	100.00%	100.00%	97.88%	92.45%	94.83%	93.62%	96.00%	Overfitting
Ada Boost	87.61%	87.47%	87.71%	87.59%	88.00%	84.26%	74.46%	52.27%	61.42%	80.00%	Overfitting
Gradient Boost	92.63%	92.46%	92.78%	92.62%	93.00%	88.98%	78.06%	64.20%	70.45%	85.00%	Overfitting
XG Boost	99.96%	99.94%	99.98%	99.96%	100.00%	97.03%	88.13%	93.87%	90.91%	93.00%	Overfitting
Tuned Models											
Models	Train					Validation					Performance
	Accuracy	Recall	Precision	F1-Score	AUC Score	Accuracy	Recall	Precision	F1-Score	AUC Score	
XG Boost - Actual	95.18%	90.44%	99.91%	94.94%	99.00%	96.13%	82.73%	93.50%	87.79%	91.00%	Overfitting
XG Boost - Oversampled	99.88%	99.81%	99.95%	99.88%	1.00%	96.43%	88.85%	89.82%	89.33%	93.00%	Overfitting
Random Forest - Actual	95.97%	94.70%	83.54%	88.77%	95.00%	93.22%	88.85%	75.30%	81.52%	91.00%	Best Fit
Random Forest - Oversampled	96.52%	96.50%	96.55%	96.52%	97.00%	92.62%	87.41%	73.64%	79.93%	91.00%	Good Model
Bagging - Actual	99.96%	99.76%	100.00%	99.88%	100.00%	97.34%	88.13%	95.70%	91.76%	94.00%	Overfitting
Bagging - Oversamp	99.98%	99.97%	100.00%	99.98%	100.00%	97.40%	92.81%	91.81%	92.31%	96.00%	Overfitting

Table 6: Model Performance

EFFORT TO IMPROVE MODEL PERFORMANCE:

The top-performing models, trained on both actual and oversampled datasets, were fine-tuned across multiple hyperparameters to enhance performance. Among these, the configurations listed below yielded the most optimal results.

RANDOM FOREST TUNED ON ACTUAL DATA:

RandomForestClassifier

```
RandomForestClassifier(class_weight='balanced', max_depth=9, max_features=0.9,
                        max_samples=0.9, n_estimators=150, random_state=1)
```

Fig 20: RF Hyperparameters

Train			Validation		
Metrics	Before	After	Metrics	Before	After
Accuracy	100.00%	95.97%	Accuracy	97.76%	93.22%
Recall	100.00%	94.70%	Recall	88.49%	88.85%
Precision	100.00%	83.54%	Precision	98.01%	75.30%
F1-Score	100.00%	88.77%	F1-Score	93.01%	81.52%
AUC Score	100.00%	95.00%	AUC Score	94.00%	91.00%

Table 7: Model improvement

BEST MODEL – RANDOM FOREST TUNED ON ACTUAL DATA:

RF Model on Actual data					
Data set	Accuracy	Recall	Precision	F1-Score	AUC Score
Train	95.97%	94.70%	83.54%	88.77%	95.00%
Validation	93.22%	88.85%	75.30%	81.52%	91.00%
Test	93.02%	89.38%	74.39%	81.20%	92.00%

Table 8: Final model performance

RF Tuned Model –Actual:

- The Random Forest model trained on actual data demonstrates consistently high recall across training (94.70%), validation (88.85%), and test (89.38%) datasets, shows its ability to detect churn cases effectively with minimal bias and strong generalization.
- The ROC AUC scores—0.95 (train), 0.91 (validation), and 0.92 (test)—confirm this model is well-suited for production.

MODEL VALIDATION:

- Given that the model is designed to predict customer churn, our primary objective is to maximize recall—ensuring we identify as many potential churn cases as possible.
- This is achieved with minimal compromise to other key performance metrics such as accuracy, precision, and F1 score.

FEATURE IMPORTANCE:

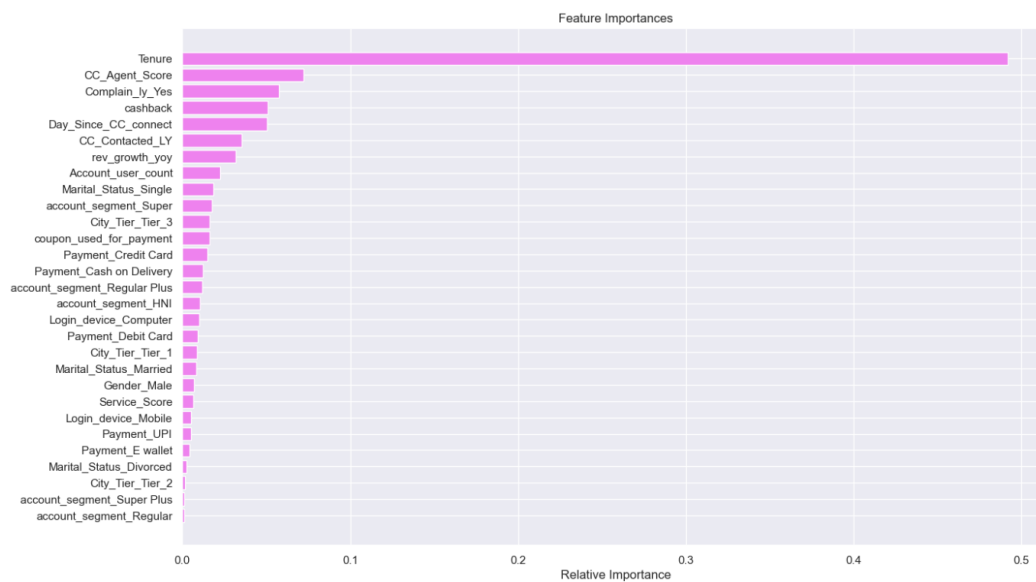


Fig 21: Important Features

Top 5 features that contribute in Customer churn classification are:

1. Tenure
2. CC_Agent_Score
3. Complain_ly_Yes
4. Cashback
5. Day_Since_CC_connect

BUSINESS INSIGHTS:

- **City Tier:** Tier 3 cities have the highest churn (21.4%), Tier 1 the lowest (14.5%).
- **Payment Method:** Cash on Delivery and E-Wallet users churn more than card users.
- **Account Segment:** Regular Plus has the most customers—but also the highest churn.
- **Marital Status:** Single customers churn more; married ones stay longer.
- **Complaints:** Customers with complains on last 12 months seems to churn the most.
- **Tenure:** Newer customers (<5 months) churn more.
- **Cashback:** Less cashback often means more churn.
 - In City Tier 1, customers who preferred Debit Cards received cashback frequently, whereas those opting for E-Wallets received none.
 - In City Tier 2, frequent cashback was observed among UPI users, with E-Wallet users again receiving none.
 - Interestingly, in City Tier 3, the pattern shifts—E-Wallet users received the most cashback, followed by those using Debit Cards.
- **Coupons:** Customers with less coupons churns often.
- **Account Structure:** Accounts with too few or too many users show higher churn.
- **Tenure Patterns:** Regular and Super Plus segments have longer tenure, especially in Tier 1 and 2 cities.
- Account segment and Tenure: 36.63% of customers chose the Regular Plus segment, their average tenure is significantly lower than other segments.
- Including City_Tier in the analysis reveals that Regular segment customers demonstrate the highest tenure across all three tiers, followed by Super Plus.
- The Regular Plus segment has the most customers, but their average tenure is low. Most of them contacted customer care just 2 or 3 days ago, which suggests frequent issues. This might also be the reason for less Tenure.
- Although only 4.62% of customers belong to the Regular segment, they tend to have a longer tenure compared to other segments. Their engagement with customer care is also relatively better.

BUSINESS RECOMMENDATIONS:

Tenure:

- Customers with a tenure of less than five months are at a higher risk of churning, especially during their early post-signup period.
- This vulnerability is often driven by competitive market offers that carry them away.
- To counter this, businesses should prioritize engagement during these crucial early months by providing frequent cashbacks, compelling promotional offers, and discounted first-year memberships such as 50% offer to strengthen retention and build loyalty.

CC_Agent_Score:

- This is customer satisfaction score reflects the quality of service delivered by customer care agents.
- To reduce the risk of churn from poor service interactions, businesses must take proactive steps to ensure top-tier support quality.
- Investing in training, monitoring service standards, and fostering empathetic communication can significantly enhance the customer experience and build long-term loyalty.

Complain_ly_Yes:

- Customers who have raised complaints in the past year require special attention.
- Businesses must actively monitor repeat issues to ensure they're not recurring.
- If the same problem persists, offering a free replacement or a full refund can demonstrate commitment to customer satisfaction helping to rebuild trust and prevent churn.

Cashback:

- Cashback is a powerful lever in influencing customer retention.
- When aligned with tenure, it can play a critical role in reducing early-stage churn.
- Providing more frequent or higher-value cashback during the initial months post-signup helps Business in reducing chances of early churn.

Day_since_CC_contact:

- Customers who have recently contacted customer care should be proactively monitored.
- Prompt acknowledgment of their concerns, followed by timely and effective resolution, is critical.
- Taking swift action not only strengthens customer trust but also reduces the likelihood of churn driven by unresolved service issues.

Account_segment:

- The Regular Plus segment appears to be the most susceptible to churn.
- Customers in this group typically exhibit shorter tenure and frequent contact with customer care, pointing to recurring service issues or unmet expectations.
- This suggests an urgent need for targeted retention strategies focused on strengthening engagement during the early lifecycle, proactively resolving service concerns, and restoring trust through tailored offers or enhanced support

Expanding Customer Base in Tier 2 and Tier 3 Cities:

Customer distribution analysis reveals a significant concentration in Tier 1 cities, while Tier 2 and Tier 3 cities remain underpenetrated specifically in Tier 2.

To boost overall customer volume and tap into emerging markets, the business should prioritize customer acquisition in these regions by performing:

- Regional Offers.
- Targeted Marketing.
- Customer Referral Programs.
- Expanding Delivery Network.

APPENDIX A:

VISUAL INSPECTION OF DATA:

DATA OVERVIEW OF FIRST AND LAST 5 ROWS WITH 12 COLUMNS:

	AccountID	Churn	Tenure	City_Tier	CC_Contacted_LY	Payment	Gender	Service_Score	Account_user_count	account_segment	CC_Agent_Score	Marital_Status
0	20000	1	4	3.00	6.00	Debit Card	Female	3.00	3	Super	2.00	Single
1	20001	1	0	1.00	8.00	UPI	Male	3.00	4	Regular Plus	3.00	Single
2	20002	1	0	1.00	30.00	Debit Card	Male	2.00	4	Regular Plus	3.00	Single
3	20003	1	0	3.00	15.00	Debit Card	Male	2.00	4	Super	5.00	Single
4	20004	1	0	1.00	12.00	Credit Card	Male	2.00	3	Regular Plus	5.00	Single

Table 9 – First 5 rows of the data

	AccountID	Churn	Tenure	City_Tier	CC_Contacted_LY	Payment	Gender	Service_Score	Account_user_count	account_segment	CC_Agent_Score	Marital_Status
11255	31255	0	10	1.00	34.00	Credit Card	Male	3.00	2	Super	1.00	Married
11256	31256	0	13	1.00	19.00	Credit Card	Male	3.00	5	HNI	5.00	Married
11257	31257	0	1	1.00	14.00	Debit Card	Male	3.00	2	Super	4.00	Married
11258	31258	0	23	3.00	11.00	Credit Card	Male	4.00	5	Super	4.00	Married
11259	31259	0	8	1.00	22.00	Credit Card	Male	3.00	2	Super	3.00	Married

Table 10 – Last 5 rows of the data

Observation:

- The dataset begins with AccountID 20000 and ends with AccountID 31259.
- The target variable for prediction is Churn.
- Some of the features contain the value 0, which needs further analysis to assess relevance or data quality.
- Missing values are present in the dataset and will need to be addressed during preprocessing.

DATA INFORMATION AFTER TREATING INCONSISTENCY:

```
RangeIndex: 11260 entries, 0 to 11259
Data columns (total 19 columns):
#   Column                Non-Null Count  Dtype
---  -
0   AccountID              11260 non-null  int64
1   Churn                  11260 non-null  int64
2   Tenure                 11042 non-null  float64
3   City_Tier              11148 non-null  object
4   CC_Contacted_LY        11158 non-null  float64
5   Payment                11151 non-null  object
6   Gender                 11152 non-null  object
7   Service_Score          11154 non-null  float64
8   Account_user_count      10816 non-null  float64
9   account_segment        11163 non-null  object
10  CC_Agent_Score          11144 non-null  float64
11  Marital_Status          11048 non-null  object
12  rev_per_month           10469 non-null  float64
13  Complains_ly            10903 non-null  object
14  rev_growth_yoy          11257 non-null  float64
15  coupon_used_for_payment 11257 non-null  float64
16  Day_Since_CC_connect    10902 non-null  float64
17  cashback                10787 non-null  float64
18  Login_device            10500 non-null  object
```

Fig 22: Data information after treating inconsistency

Observation:

- The following columns contained inconsistent data types and have been cleaned and converted to float:
 - Tenure
 - Account_user_count
 - rev_per_month
 - rev_growth_yoy
 - coupon_used_for_payment
 - Day_Since_CC_connect
 - cashback
- Missing values in some columns increased due to the replacement of special characters with NaN.
- Values in the Gender column were standardized by converting 'M' and 'F' to 'Male' and 'Female', respectively.
- Similarly values in the account_segment column were standardized by converting 'Regular +' and 'Super +' to Regular Plus and Super Plus respectively.

Note:

Churn column has been converted as Object for analysis purpose.

Random 10 rows after the initial clean up:

	Churn	Tenure	City_Tier	CC_Contacted_LY	Payment	Gender	Service_Score	Account_user_count	account_segment	CC_Agent_Score	Marital_Status
784	0	0.00	1.00	31.00	Debit Card	Female	2.00	1.00	Regular Plus	2.00	Single
6943	1	0.00	3.00	22.00	E wallet	Male	2.00	3.00	Regular Plus	5.00	Married
3709	0	10.00	3.00	23.00	E wallet	Female	4.00	5.00	Super	5.00	Divorced
6439	0	0.00	1.00	10.00	Credit Card	Female	2.00	3.00	Regular Plus	4.00	Single
5310	0	9.00	1.00	25.00	Debit Card	Male	3.00	2.00	Super	5.00	Married
6964	0	2.00	1.00	21.00	UPI	Male	2.00	4.00	Super	4.00	Married
2333	0	14.00	1.00	28.00	Debit Card	Female	3.00	3.00	Super	1.00	Married
7315	0	10.00	1.00	13.00	Credit Card	Male	3.00	NaN	Super	1.00	Married
10299	0	7.00	1.00	16.00	Debit Card	Male	2.00	4.00	Super	1.00	Married
10411	0	10.00	3.00	10.00	Debit Card	Male	3.00	4.00	Super	1.00	Married

Table 10 – Random 10 rows of the data

NUMERIC COLUMNS:

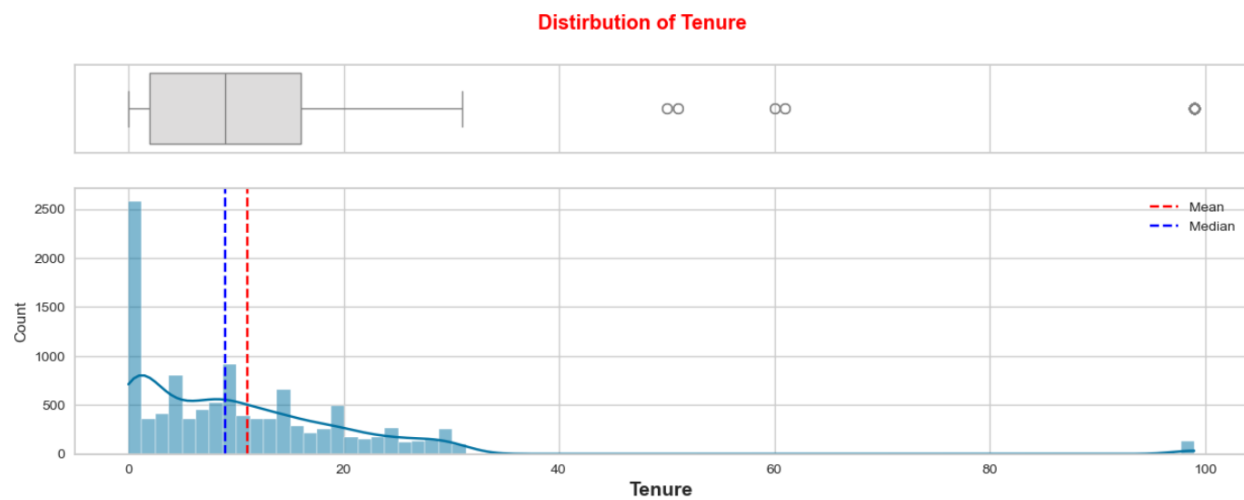


Fig 22: Distribution of Tenure

- Tenure after 40 months are outliers.
- Data is right skewed since mean is greater than median.

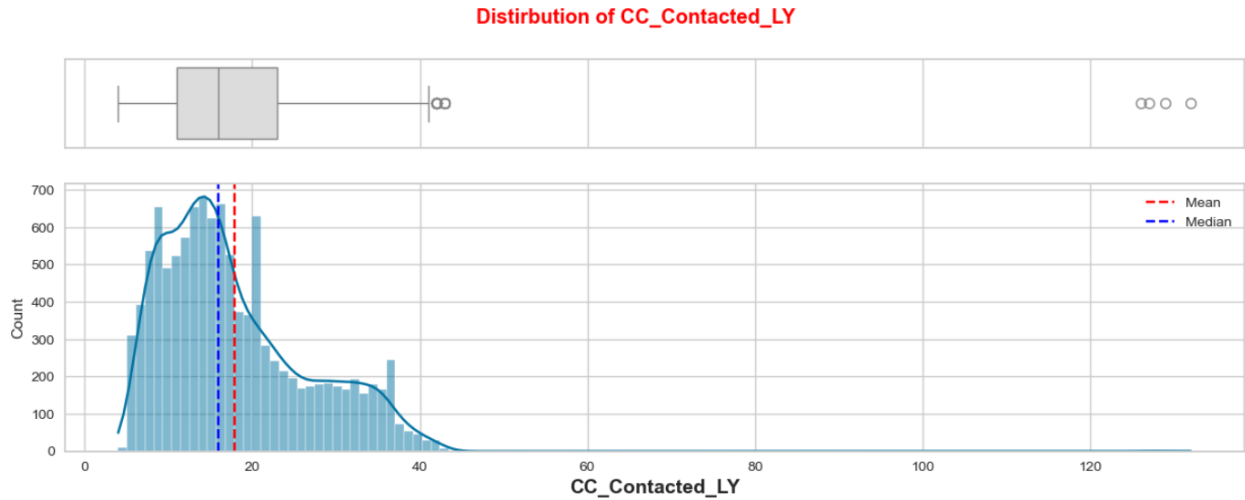


Fig 23: Distribution of CC_Contacted_LY

- Data is right skewed since mean is greater than median.

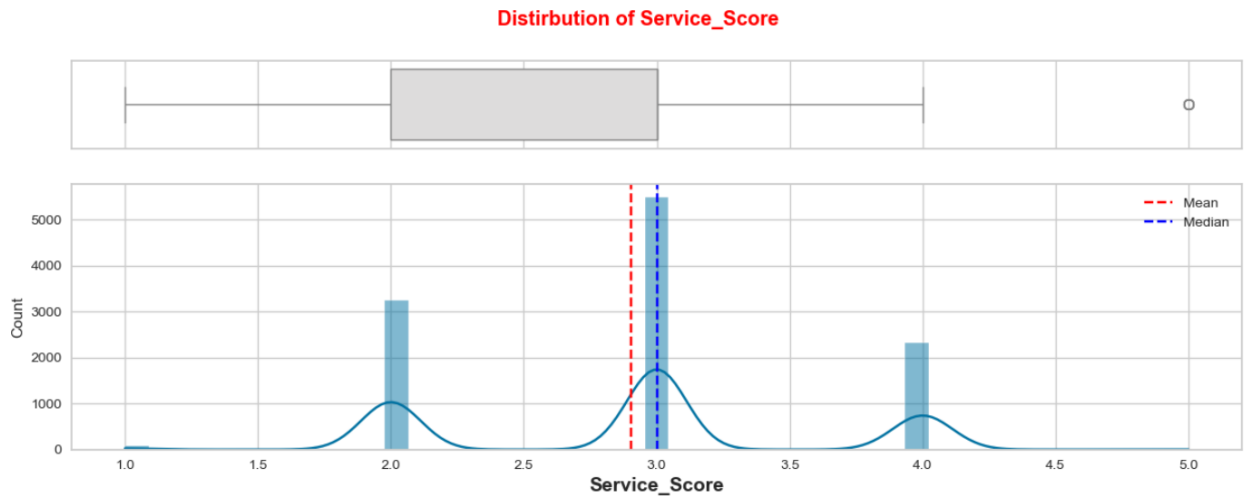


Fig 24: Distribution of Service_score

- The majority of customers rated their service experience as 3, followed by ratings of 2 and 4. A rating of 5 appears to be an outlier, indicating very few customers selected the highest score.

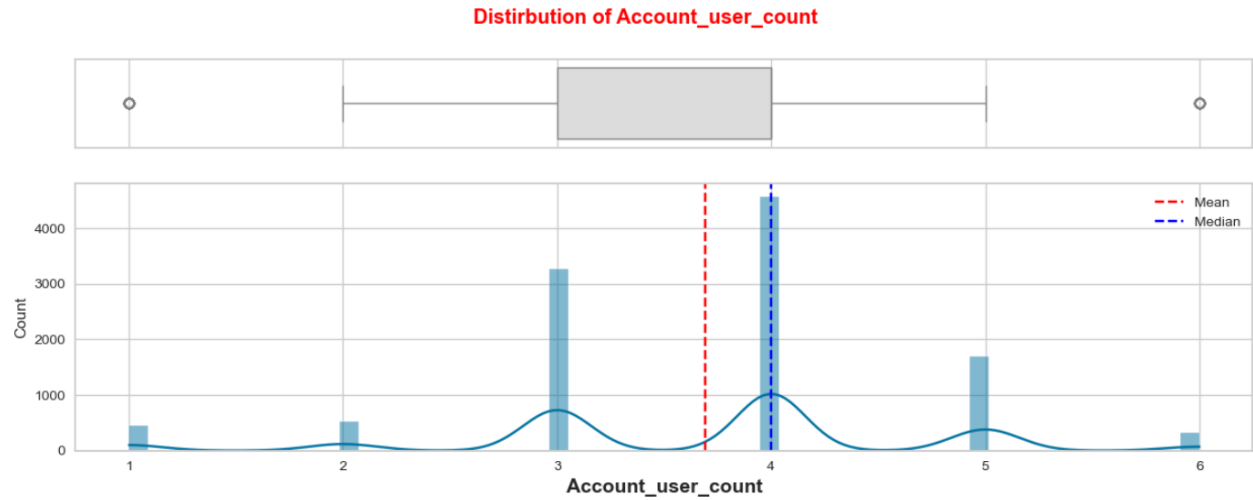


Fig 25: Distribution of Account_user_count

- Most accounts have 4 customers tagged, followed by those with 3 and 5 customers. Accounts with 1 or 6 customers are rare and considered outliers.

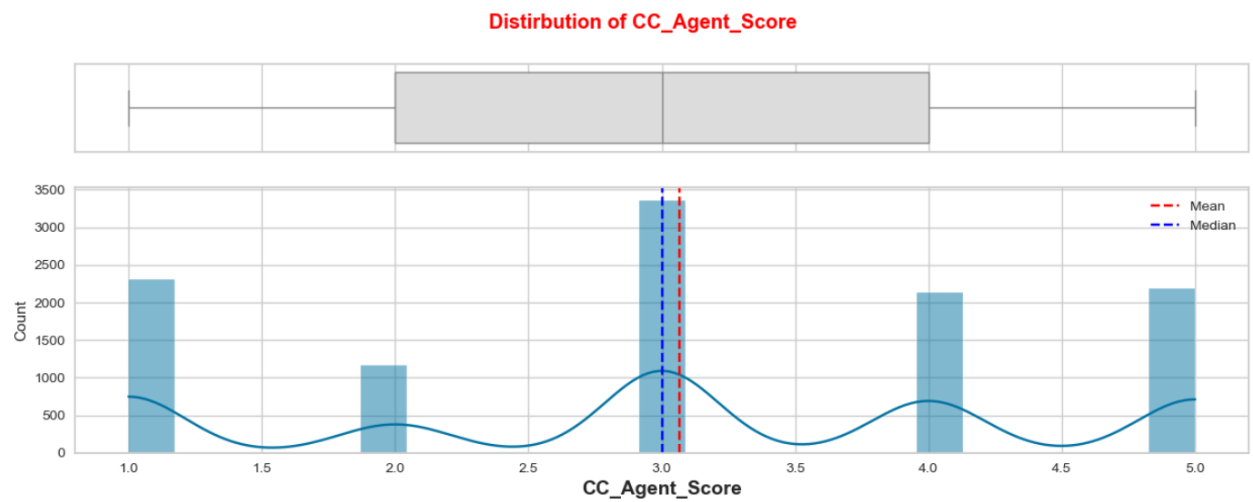


Fig 26: Distribution of CC_Agent_Score

- The data appears to be free of outliers.

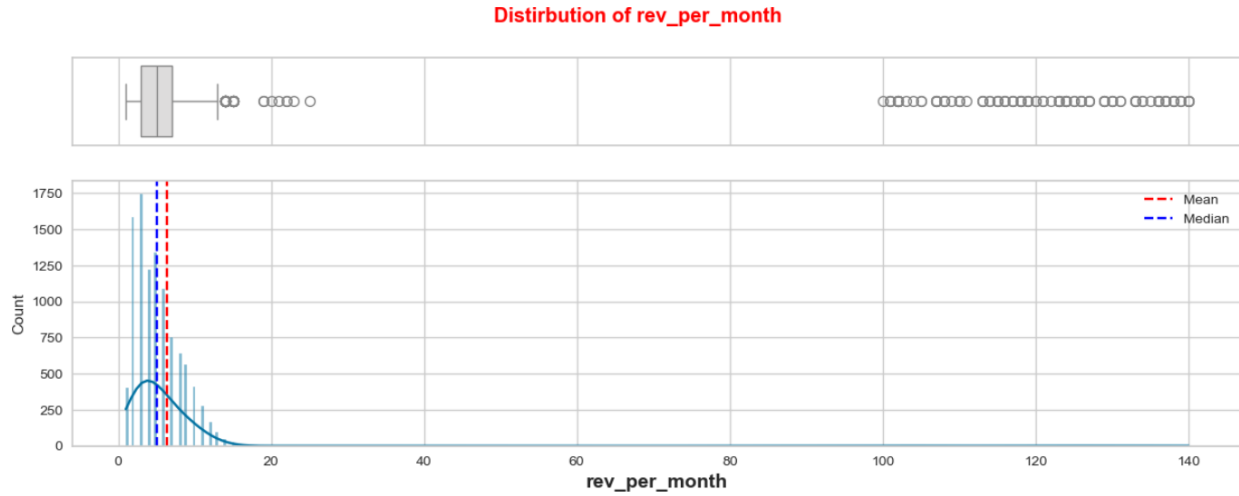


Fig 27: Distribution of rev_per_month

- The data contains numerous outliers.
- Additionally, since the mean is greater than the median, the data distribution is right-skewed.

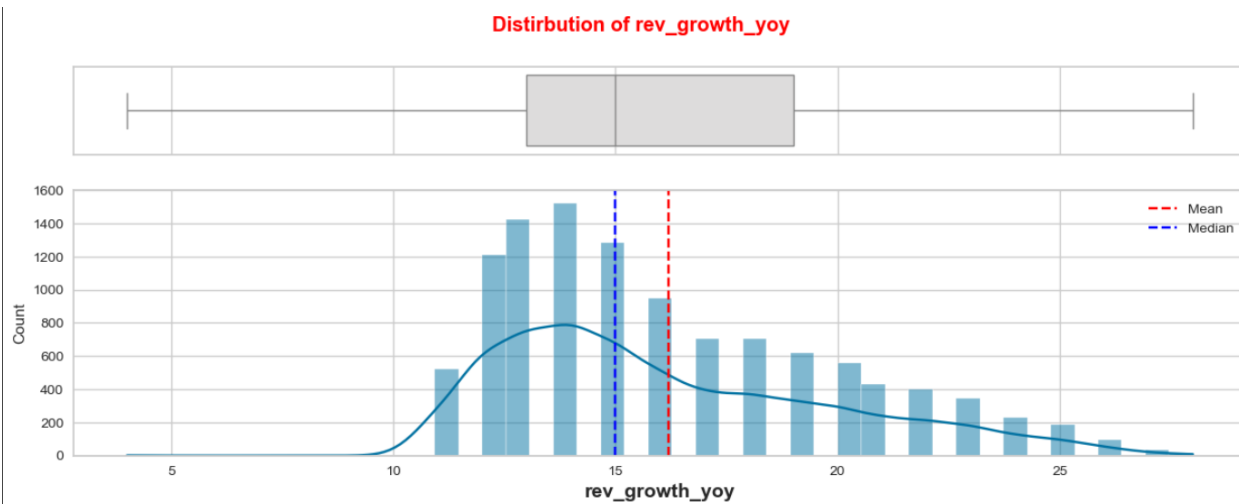


Fig 28: Distribution of rev_growth_yoy

- The data appears to be free of outliers.
- The mean is greater than the median, the data distribution is right-skewed.

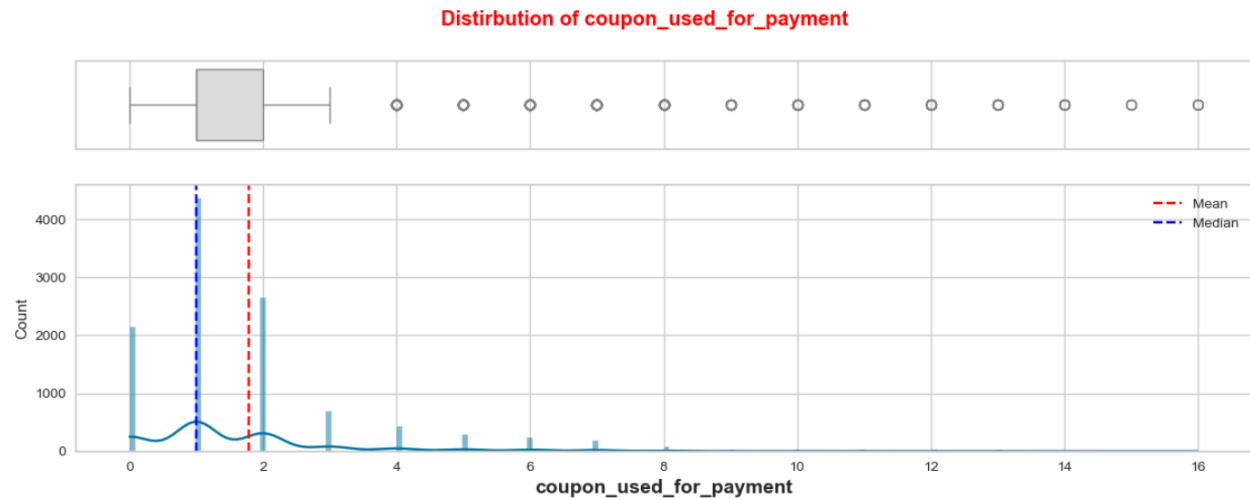


Fig 29: Distribution of coupon_used_for_payment

- This data has few outliers.
- Since the mean exceeds the median, the data distribution is right-skewed.

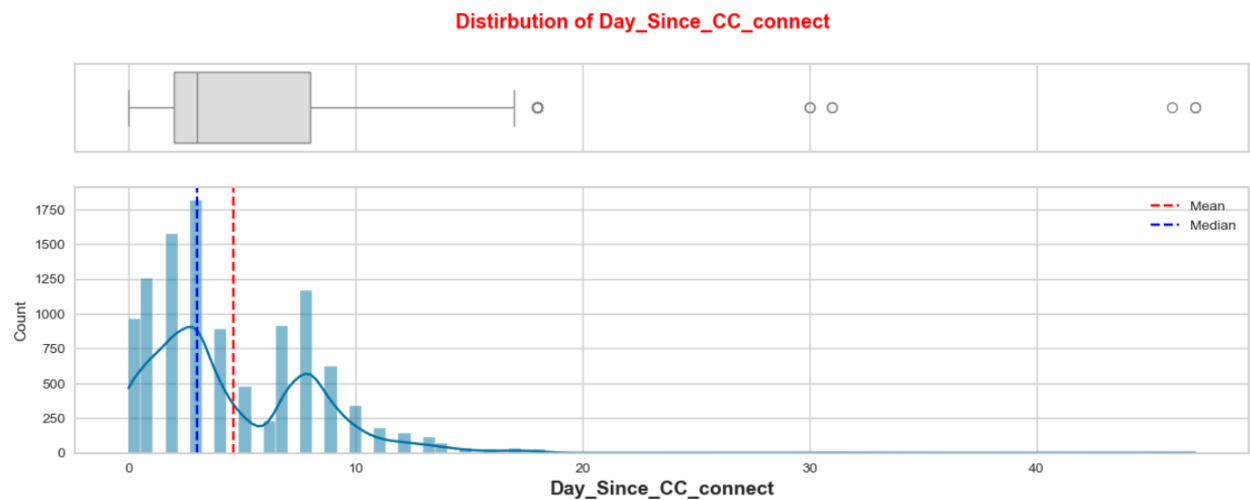


Fig 30: Distribution of Day_Since_CC_Connect

- This data has few outliers.
- Since the mean exceeds the median, the data distribution is right-skewed.

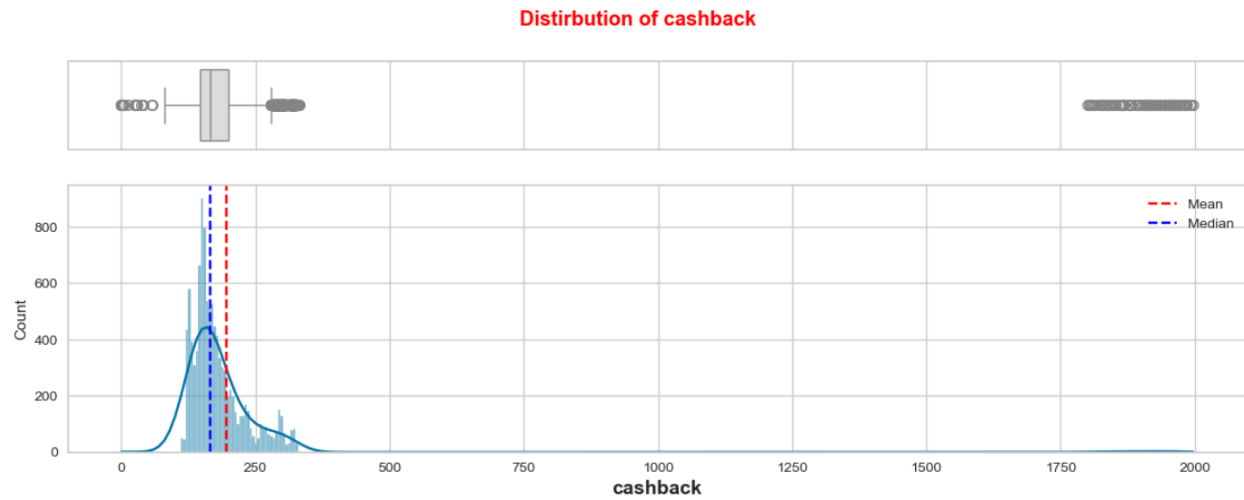


Fig 31: Distribution of cashback

- This data has many outliers.
- Since the mean exceeds the median, the data distribution is right-skewed.

APPENDIX B

MODELS:

1.LOGISTIC REGRESSION:

	LR_Train	LR_Validation
Accuracy	89.59	90.44
Recall	55.49	56.47
Precision	76.14	80.93
F1	64.20	66.53

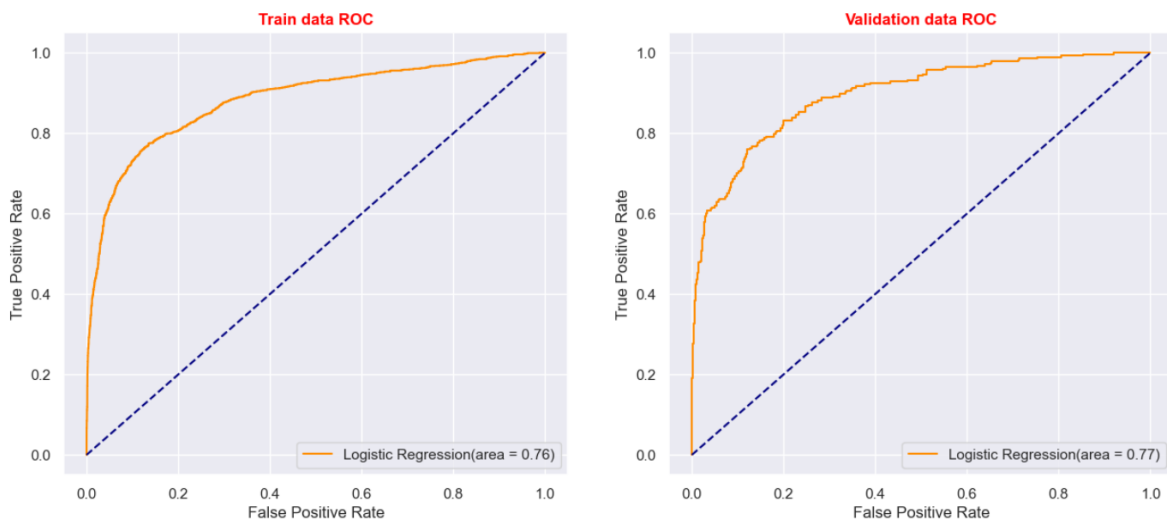


Fig 32: Logistic regression AUC_ROC

- At a threshold of 0.5, logistic regression yields a training recall of 55.49%, suggesting high bias and limited learning of underlying patterns.
- On the validation set, it maintains a recall of 56.47%, indicating low variance and better generalization.
- However, the overall performance suggests underfitting, as the model struggles to fully capture the complexity of the data.
- Across 5-fold cross-validation, average recall is 56%, almost similar to train recall.
- This model achieves a ROC AUC of 0.76 in train and 0.77 in validation, which is decent but there's still room for improvement.

2.NAIVE BAYES:

	NB_Train	NB_Validation
Accuracy	80.98	80.99
Recall	70.83	73.38
Precision	45.76	45.95
F1	55.60	56.51

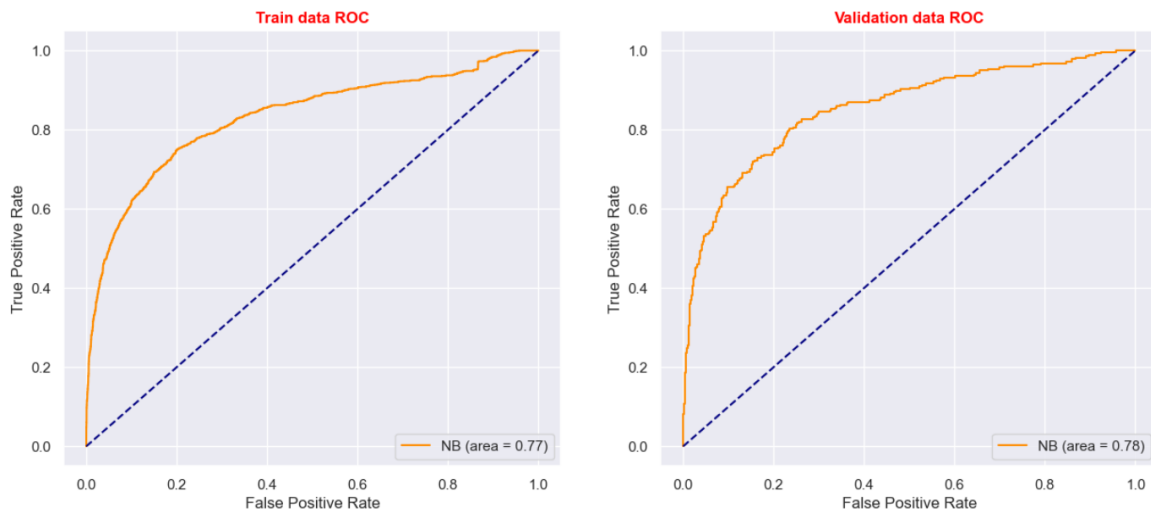


Fig 33: Naïve Bayes AUC_ROC

- The Naive Bayes model outperformed logistic regression, delivering a training recall of 70.83%, which suggests moderate bias, the model learns underlying patterns better but still simplifies relationships.
- On the validation set, it achieved a recall of 73.38%, reflecting better variance and robust generalization.
- Overall performance is decent, though not optimal.
- Across 5-fold cross-validation, the model shows consistent recall averaging 71%, closely aligned with the training score
- This model achieves a ROC AUC of 0.77 on train and 0.78 on validation, which is good but there's still room for improvement.

3.LDA:

	LDA_Train	LDA_Validation
Accuracy	89.72	90.19
Recall	60.16	59.35
Precision	73.88	77.10
F1	66.32	67.07

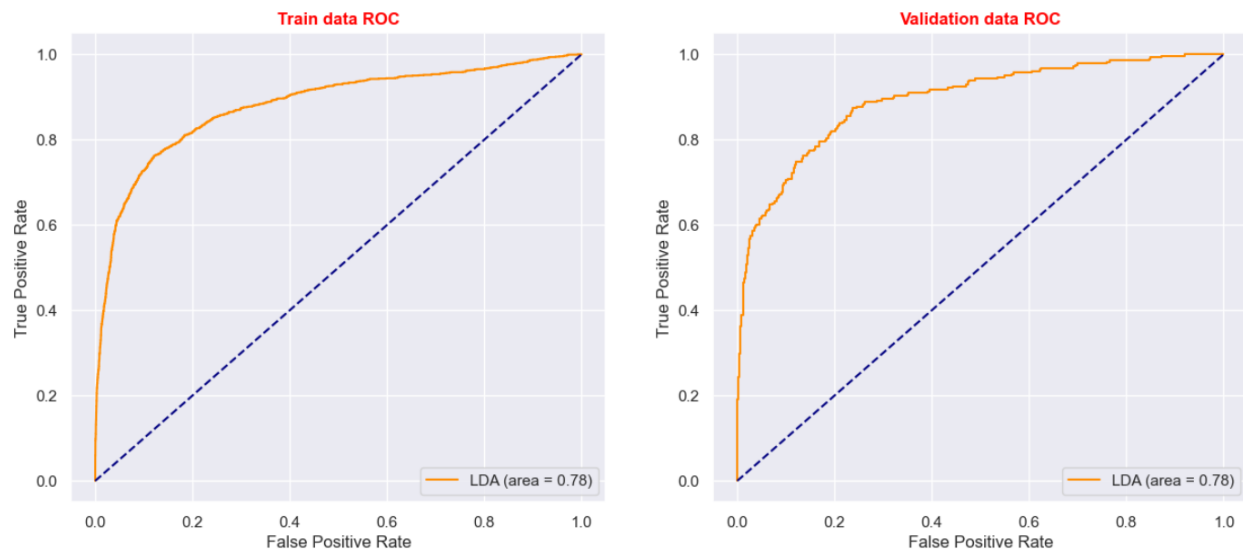


Fig 34: LDA AUC_ROC

- The model achieves a training recall of 60.16%, indicating high bias and limited capacity to capture complex patterns.
- Its validation recall of 59.35% suggests low variance and stable generalization on unseen data.
- However, the overall performance suggests underfitting, as the model struggles to fully capture the complexity of the data.
- Across 5-fold cross-validation, the model shows consistent recall averaging 60%, closely aligned with the training score
- It records a ROC AUC of 0.78 on train and 0.78 on validation, which is good but there's still room for improvement.

4.KNN:

	KNN_Train	KNN_Validation
Accuracy	98.06	95.22
Recall	91.70	77.70
Precision	96.59	92.70
F1	94.08	84.54

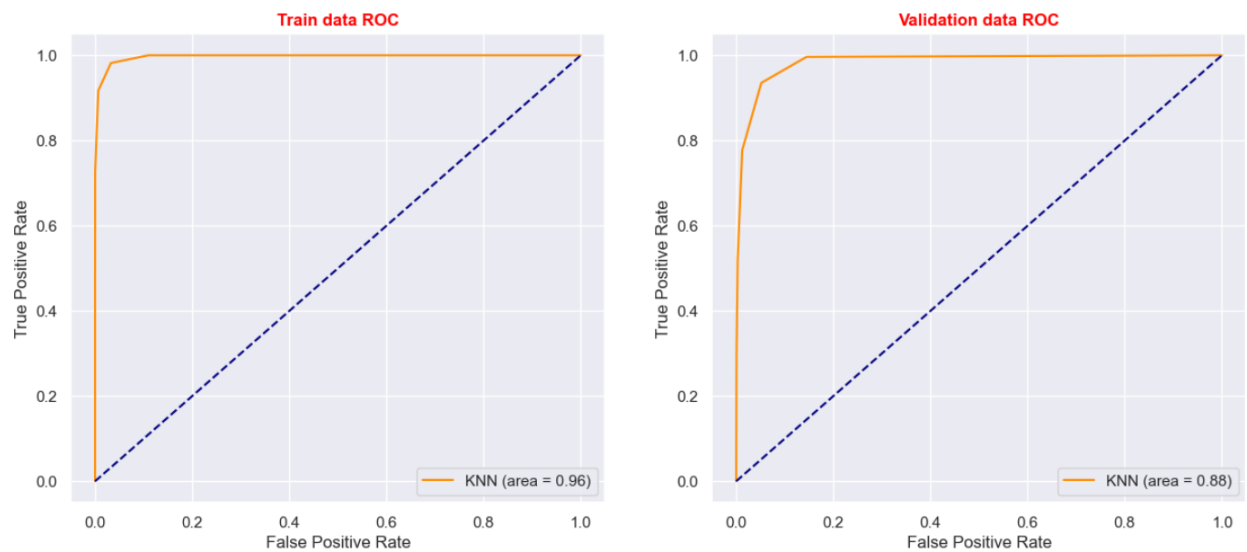


Fig 35: KNN AUC_ROC

- The model delivers a training recall of 91.70%, indicating low bias and strong pattern learning on known data.
- On the validation set, recall drops to 77.70%, which suggests high variance and poor generalization performance.
- This Model is Overfitting in nature.
- However, the cross-validation recall average of 73%, notably lower than the training score, reveals high variance across folds proves overfitting nature.
- It records a ROC AUC of 0.96 on train and 0.88 on validation, which is again proves Overfitting nature.

5.DECISION TREE:

	DT_Train	DT_Validation
Accuracy	100.00	96.49
Recall	100.00	88.13
Precision	100.00	90.74
F1	100.00	89.42

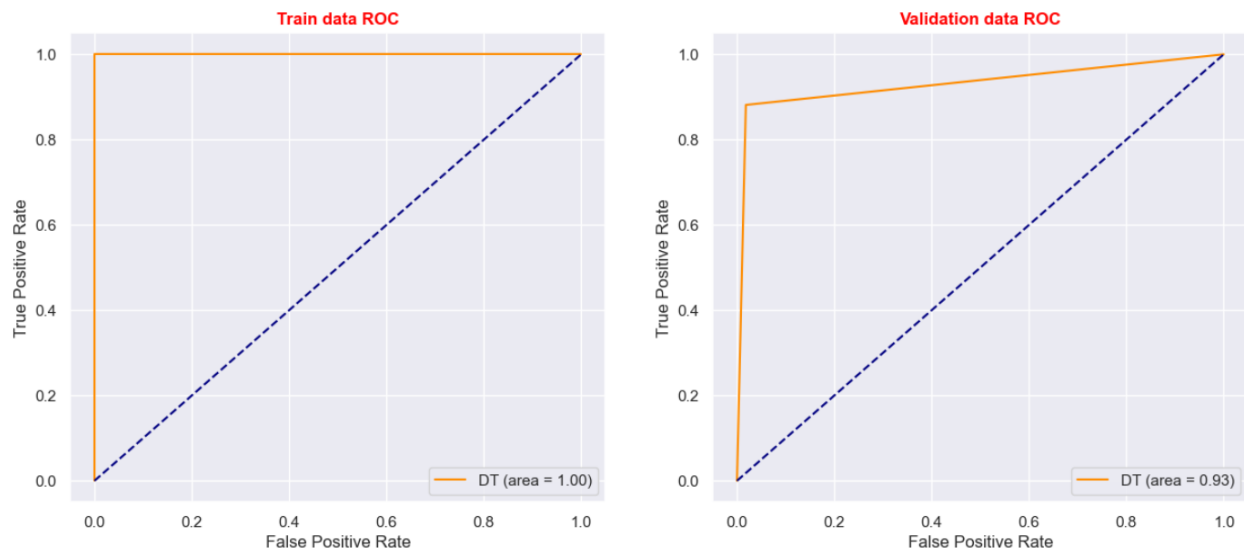


Fig 36: Decision Tree AUC_ROC

- The model delivers a training recall of 100.00%, indicating Low bias and strong pattern learning on known data.
- On the validation set, recall drops to 88.13%, which suggests high variance and weak generalization performance.
- This Model is Overfitting in nature.
- However, the cross-validation recall average of 86%, notably lower than the training score, reveals high variance across folds a key signal of potential overfitting.
- A perfect ROC AUC of 1.00 on the training set suggests the model may be overfitting.

6. BAGGING:

	Bag_Train	Bag_Validation
Accuracy	99.79	97.58
Recall	98.81	89.57
Precision	99.92	95.77
F1	99.36	92.57

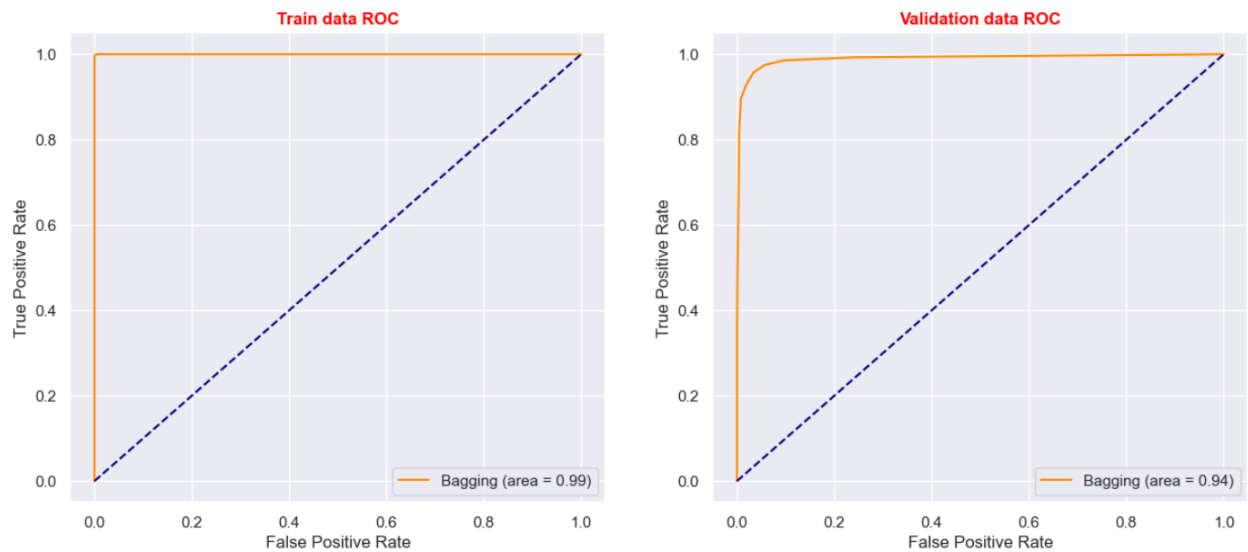


Fig 37: Bagging AUC_ROC

- The model delivers a training recall of 98.81%, indicating Low bias and strong pattern learning on known data.
- On the validation set, recall drops to 89.57%, which suggests moderate variance and moderate generalization performance.
- However, the cross-validation recall average of 82%, notably lower than the training score, reveals high variance across folds a key signal of potential overfitting.
- Train ROC AUC of 0.99 and validation ROC of 0.94 indicate strong performance overall, but the near-perfect train score raises concerns about potential overfitting.

7.RANDOM FOREST:

	RF_Train	RF_Validation
Accuracy	100.00	97.76
Recall	100.00	88.49
Precision	100.00	98.01
F1	100.00	93.01

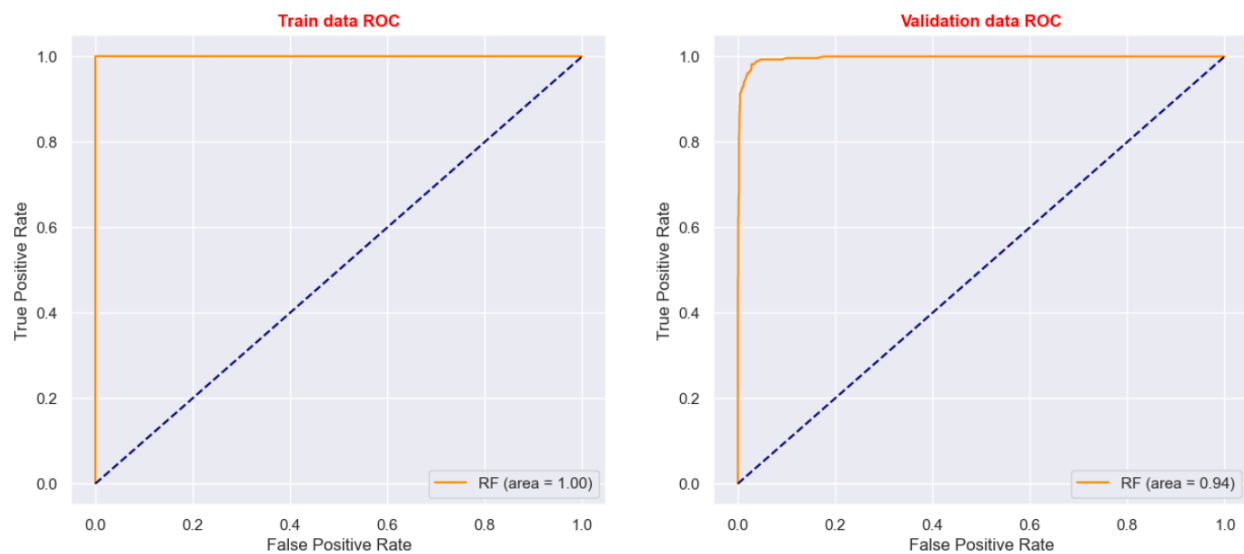


Fig 38: Random Forest AUC_ROC

- The model delivers a training recall of 100.00%, indicating Low bias and strong pattern learning on known data.
- On the validation set, recall drops to 88.49%, which suggests moderate variance and moderate generalization performance.
- However, the cross-validation recall average of 83%, notably lower than the training score, reveals high variance across folds a key signal of potential overfitting.
- A perfect ROC AUC of 1.00 on the training set suggests the model may be overfitting.

8: ADA BOOST:

	AB_Train	AB_Validation
Accuracy	89.07	88.92
Recall	56.36	56.83
Precision	72.53	71.49
F1	63.43	63.33

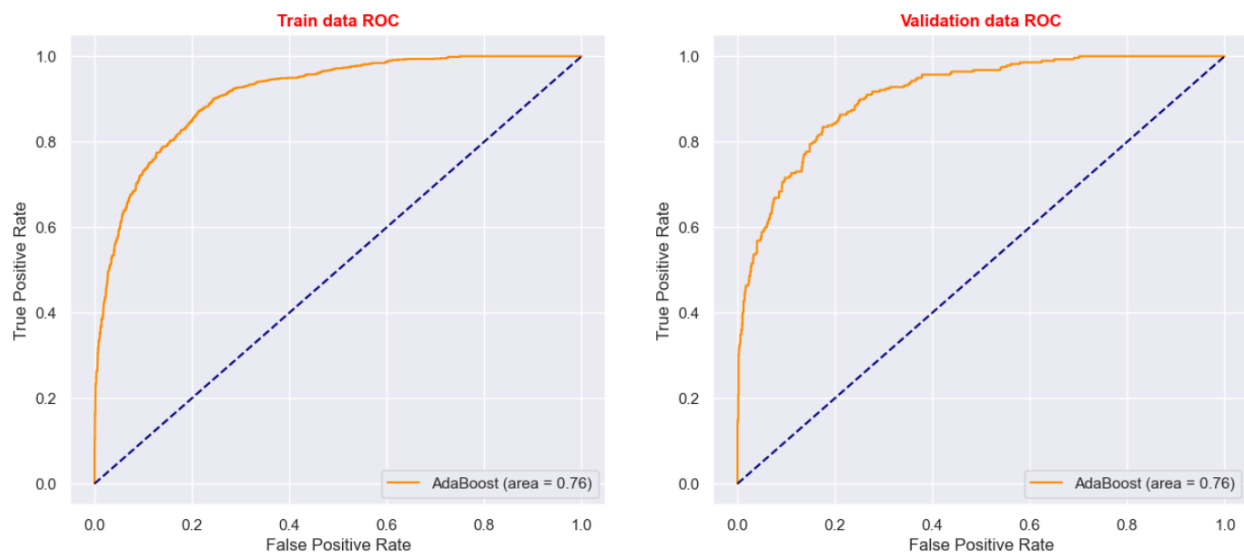


Fig 39: AdaBoost AUC_ROC

- The model delivers a training recall of 56.36%, indicating High bias and weak pattern learning on known data.
- On the validation set, recall drops to 56.83%, which suggests low variance and good generalization performance.
- However, the cross-validation recall average of 56%, same as training score, reveals low variance across folds.
- This model shows the characteristics of underfitting.
- Despite this, the ROC AUC scores for both training and validation are moderate.

9. GRADIENT BOOSTING:

	GB_Train	GB_Validation
Accuracy	91.84	91.28
Recall	62.85	60.43
Precision	84.66	83.17
F1	72.14	70.00

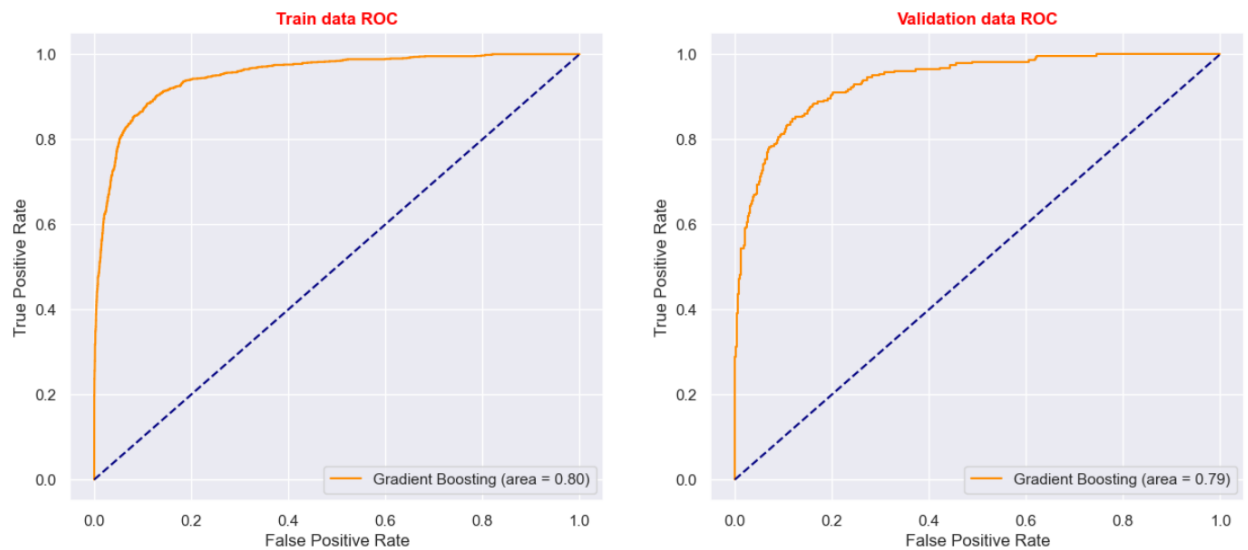


Fig 40: Gradient Boost AUC_ROC

- The model delivers a training recall of 62.65%, indicating High bias and weak pattern learning on known data.
- On the validation set, recall drops to 60.43%, which suggests low variance and good generalization performance.
- However, the cross-validation recall average of 60%, very close to training score, reveals low variance across folds.
- This model shows the characteristics of underfitting.
- Despite this, the ROC AUC scores for both training and validation are moderate.

10. XGBOOST:

	XGB_Train	XGB_Validation
Accuracy	99.99	97.46
Recall	99.92	89.93
Precision	100.00	94.70
F1	99.96	92.25

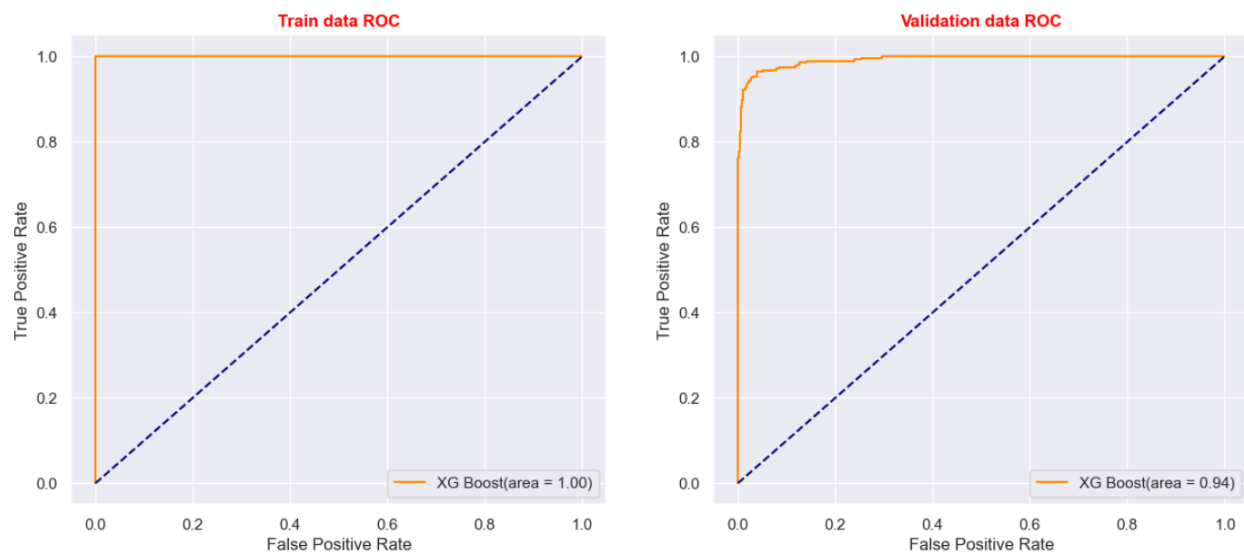


Fig 41: XGBoost AUC_ROC

- The model delivers a training recall of 99.92%, indicating Low bias and strong pattern learning on known data.
- On the validation set, recall drops to 89.93%, which suggests moderate variance and moderate generalization performance.
- However, the cross-validation recall average of 86%, notably lower than the training score, reveals high variance across folds a key signal of potential overfitting.
- A perfect ROC AUC of 1.00 on the training set suggests the model may be overfitting.

COMPAIRING RESULTS OF MODELS ON ACTUAL DATA:

	Accuracy	Recall	Precision	F1
LR_Train	89.59	55.49	76.14	64.20
NB_Train	80.98	70.83	45.76	55.60
LDA_Train	89.72	60.16	73.88	66.32
KNN_Train	98.06	91.70	96.59	94.08
DT_Train	100.00	100.00	100.00	100.00
Bag_Train	99.79	98.81	99.92	99.36
RF_Train	100.00	100.00	100.00	100.00
AB_Train	89.07	56.36	72.53	63.43
GB_Train	91.84	62.85	84.66	72.14
XGB_Train	99.99	99.92	100.00	99.96

	Accuracy	Recall	Precision	F1
LR_Validation	90.44	56.47	80.93	66.53
NB_Validation	80.99	73.38	45.95	56.51
LDA_Validation	90.19	59.35	77.10	67.07
KNN_Validation	95.22	77.70	92.70	84.54
DT_Validation	96.49	88.13	90.74	89.42
Bag_Validation	97.58	89.57	95.77	92.57
RF_Validation	97.76	88.49	98.01	93.01
AB_Validation	88.92	56.83	71.49	63.33
GB_Validation	91.28	60.43	83.17	70.00
XGB_Validation	97.46	89.93	94.70	92.25

Among these 10 models:

Underfitting models:

Logistic regression, Naïve Bayes ,LDA , AdaBoost, Gradient boosting.

Overfitting:

KNN , Decision Trees, Bagging , Random Forest and XG Boost.

Next steps:

XGBoost, Random Forest, and Bagging can be further tuned at a later stage to explore potential performance improvements.

MODELS ON OVERSAMPLED DATA:

1. LOGISTIC REGRESSION:

	LR_oversamp_Train	LR_oversamp_Validation
Accuracy	80.45	80.57
Recall	80.12	81.29
Precision	80.65	45.66
F1	80.38	58.47

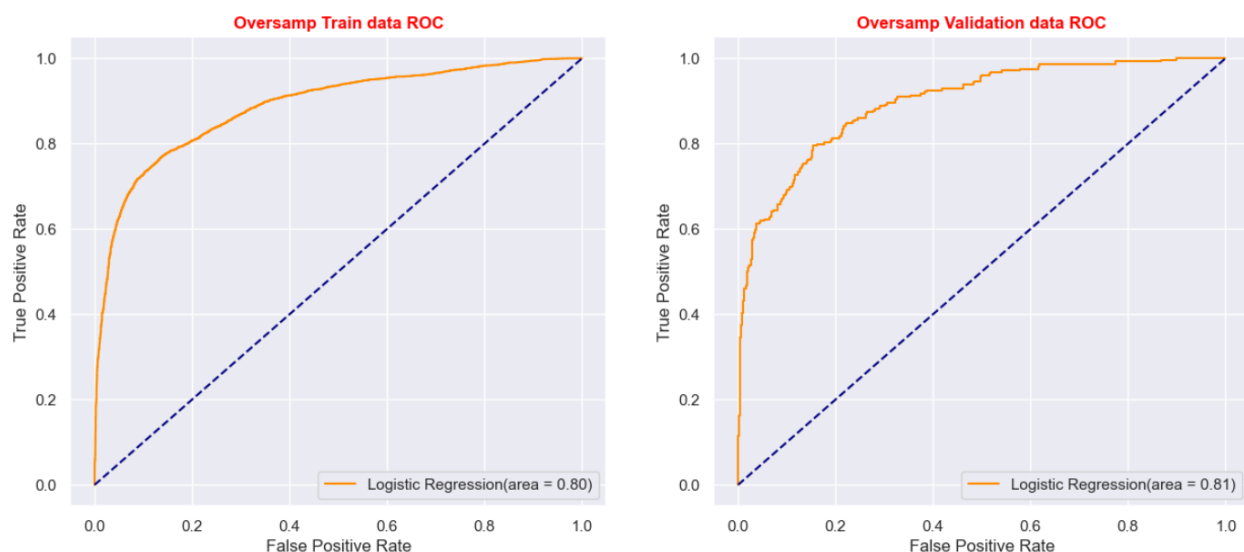


Fig 42: Logistic Regression AUC_ROC oversampled data

- At a threshold of 0.5, logistic regression applied to the oversampled dataset achieves a training recall of 80.12%, indicating moderate bias.
- On the validation set, recall holds steady at 80.57%, suggesting low variance and strong generalization.
- Overall, the model demonstrates balanced performance, though there's clear scope for improvement.
- Across 5-fold cross-validation, the average recall is 80%, closely aligned with training metrics.
- A ROC AUC of 0.83 on the training set and 0.81 on the validation set indicates the model generalizes well across datasets.

2. NAIVE BAYES:

	NB_oversamp_Train	NB_oversamp_Validation
Accuracy	74.81	67.19
Recall	84.56	85.25
Precision	70.76	32.11
F1	77.05	46.65

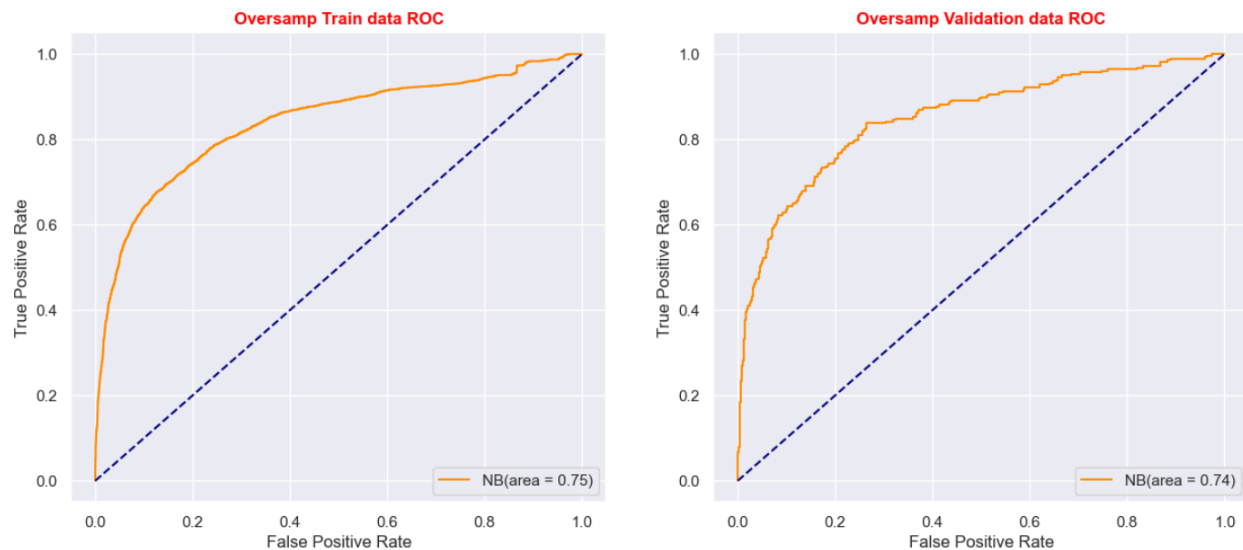


Fig 43: Naïve Bayes AUC_ROC oversampled data

- Naive Bayes on oversampled dataset achieves a training recall of 84.56%, indicating moderate bias.
- On the validation set, recall holds steady at 85.25%, suggesting low variance and strong generalization.
- Overall, the model demonstrates balanced performance, though there's clear scope for improvement.
- Across 5-fold cross-validation, the average recall is 84%, closely aligned with training metrics.
- A ROC AUC of 0.75 on the training set and 0.74 on the validation set indicates the model generalizes well across datasets but has scope of improvement.

3.LDA:

	LDA_oversamp_Train	LDA_oversamp_Validation
Accuracy	80.47	80.39
Recall	79.98	81.29
Precision	80.78	45.38
F1	80.38	58.25

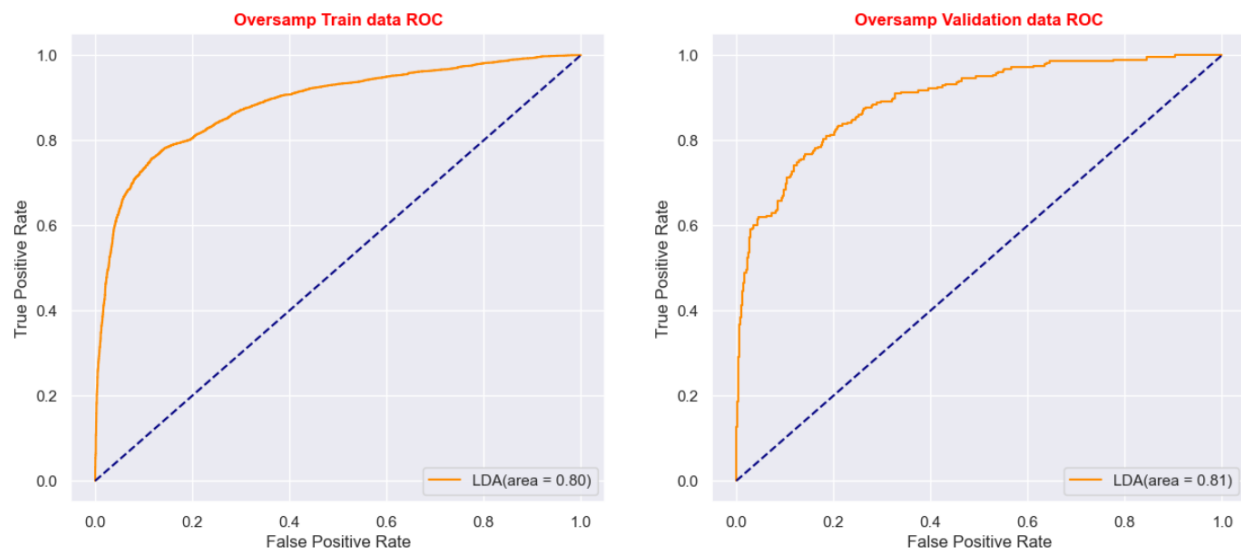


Fig 44: LDA AUC_ROC oversampled data

- LDA on oversampled dataset achieves a training recall of 79.98%, indicating moderate bias.
- On the validation set, recall holds steady at 81.29%, suggesting low variance and strong generalization.
- Overall, the model demonstrates balanced performance, though there's clear scope for improvement.
- Across 5-fold cross-validation, the average recall is 80%, closely aligned with training metrics.
- A ROC AUC of 0.80 on the training set and 0.81 on the validation set indicates the model generalizes well across datasets but has scope of improvement.

4.KNN:

	KNN_oversamp_Train	KNN_oversamp_Validation
Accuracy	98.35	93.83
Recall	99.98	98.56
Precision	96.81	73.66
F1	98.37	84.31

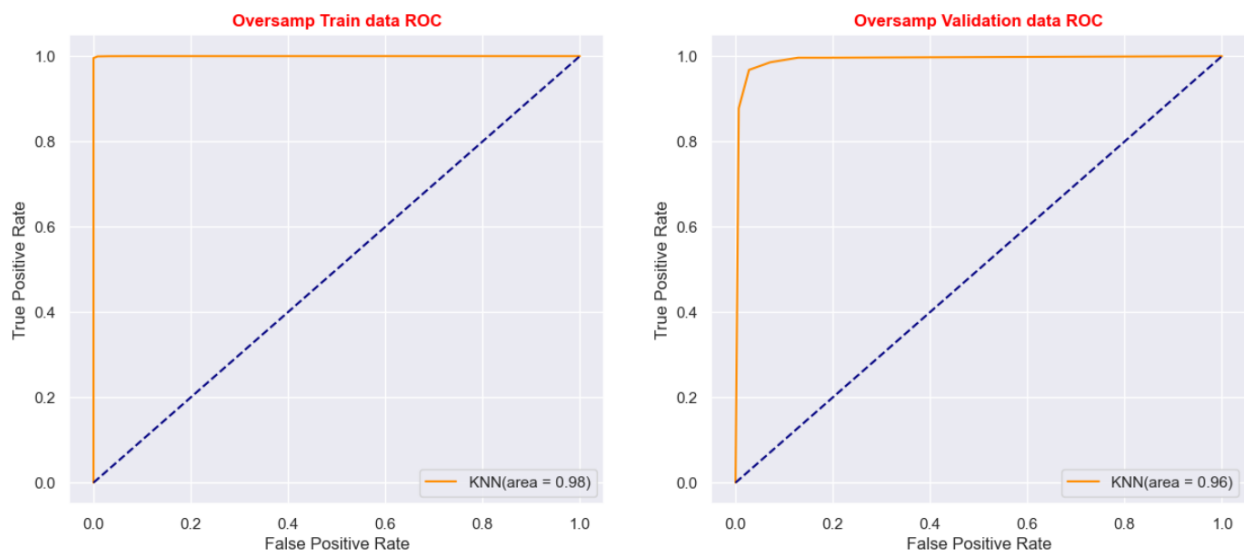


Fig 45: KNN AUC_ROC oversampled data

- The model achieves a training recall of 99.98%, suggesting very low bias, and a validation recall of 98.56%, indicating minimal variance it appears to generalize well.
- However, the cross-validation average recall is a perfect 1.0 across all five folds, which hints at potential overfitting.
- A ROC AUC of 0.98 on train and 0.96 on validation points to strong class separation, though the metrics.

5. DECISION TREE:

	DT_oversamp_Train	DT_oversamp_Validation
Accuracy	100.00	93.95
Recall	100.00	86.69
Precision	100.00	79.28
F1	100.00	82.82

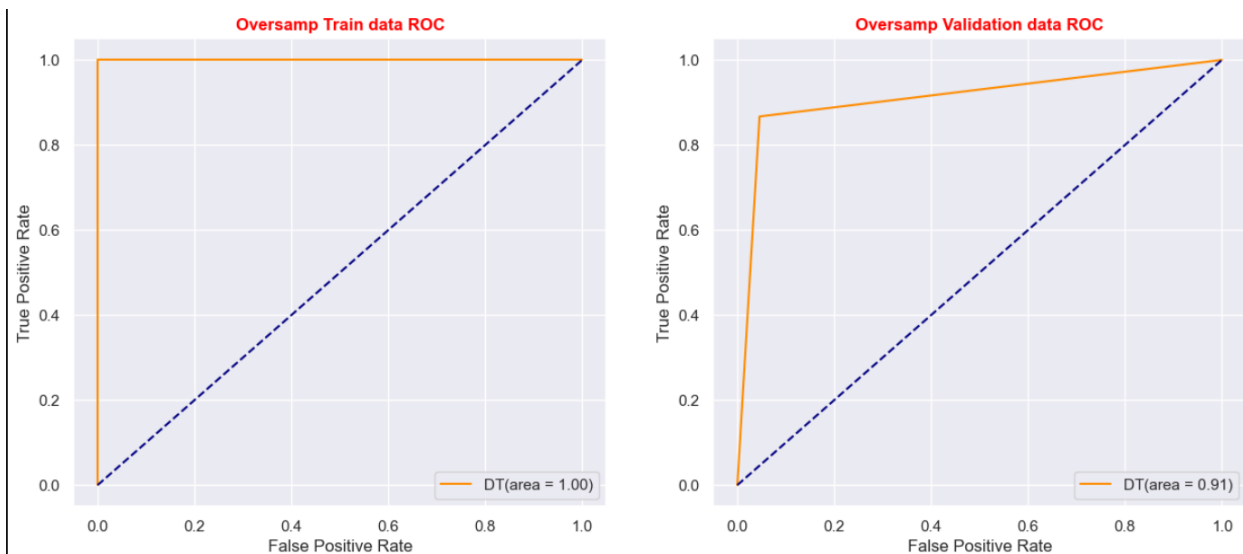


Fig 46: Decision Tree AUC_ROC oversampled data

- The model delivers a training recall of 100.00%, indicating Low bias and strong pattern learning on known data.
- On the validation set, recall drops to 86.69%, which suggests moderate variance and moderate generalization performance.
- However, the cross-validation recall average of 97%, notably lower than the training score, reveals low variance across folds.
- A perfect ROC AUC of 1.00 on the training set suggests the model may be overfitting.

6. BAGGING:

	Bag_oversamp_Train	Bag_oversamp_Validation
Accuracy	99.94	95.70
Recall	99.90	87.41
Precision	99.97	87.10
F1	99.94	87.25

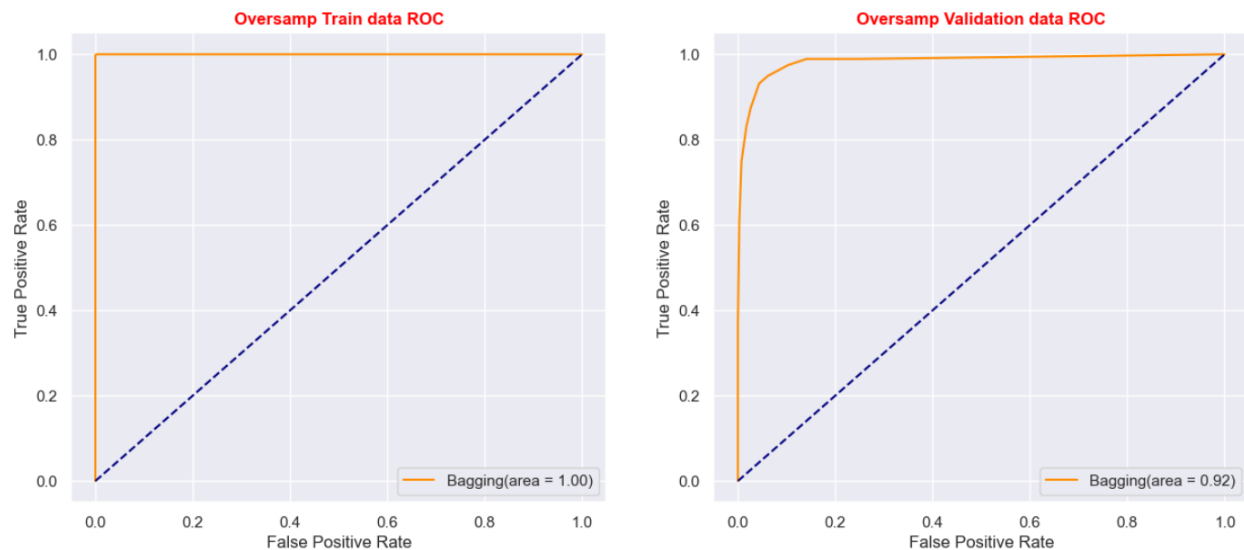


Fig 47: Bagging AUC_ROC oversampled data

- The model delivers a training recall of 99.90%, indicating Low bias and strong pattern learning on known data.
- On the validation set, recall drops to 87.41%, which suggests moderate variance and moderate generalization performance.
- However, the cross-validation recall average of 98%, notably lower than the training score, reveals low variance across folds.
- A perfect ROC AUC of 1.00 on the training set suggests the model may be overfitting.

7.RANDOM FOREST:

	RF_oversamp_Train	RF_oversamp_Validation
Accuracy	100.00	97.88
Recall	100.00	92.45
Precision	100.00	94.83
F1	100.00	93.62

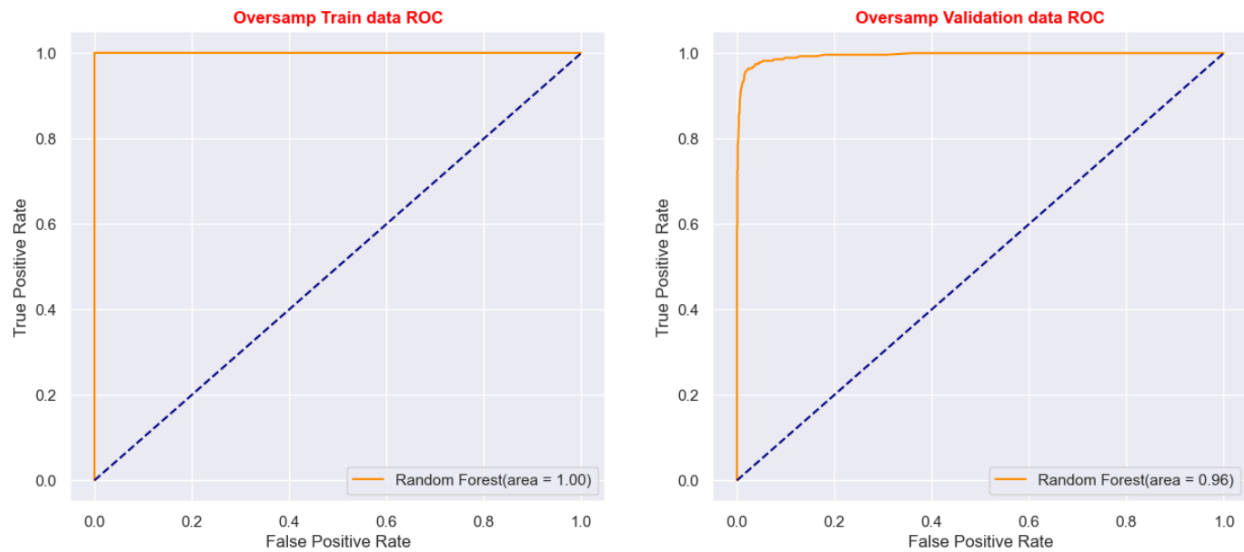


Fig 48: Random Forest AUC_ROC oversampled data

- The model delivers a training recall of 100.00%, indicating Low bias and strong pattern learning on known data.
- On the validation set, recall drops to 92.45%, which suggests low variance and good generalization performance.
- However, the cross-validation recall average of 99%, notably lower than the training score, reveals low variance across folds.
- A perfect ROC AUC of 1.00 on the training set suggests the model may be overfitting.

8 ADA BOOST:

	AB_Train	AB_Validation
Accuracy	87.61	84.26
Recall	87.47	74.46
Precision	87.71	52.27
F1	87.59	61.42

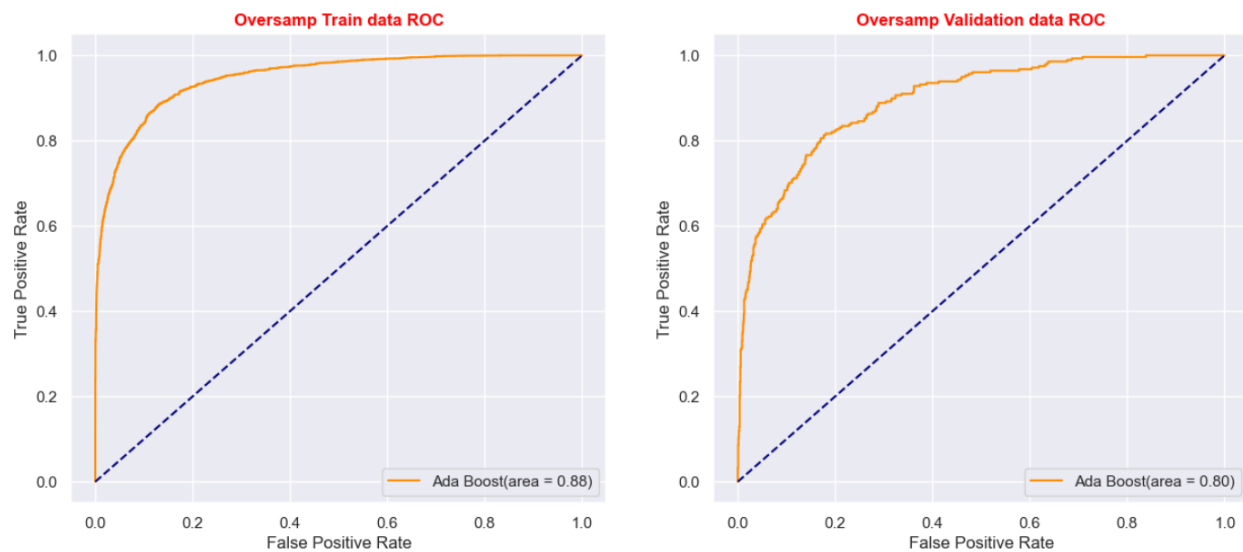


Fig 49: AdaBoost AUC_ROC oversampled data

- The model delivers a training recall of 87.47%, indicating moderate bias and moderate pattern learning on known data.
- On the validation set, recall drops to 74.46%, which suggests high variance and poor generalization performance.
- However, the cross-validation recall average of 87%, notably same as training score, reveals low variance across folds.
- A ROC AUC of 0.88 on the training set and 0.80 on the validation set indicates the model generalizes well across datasets but has scope of improvement.

9 GRADIENT BOOSTING:

	GB_oversamp_Train	GB_oversamp_Validation
Accuracy	92.63	88.98
Recall	92.46	78.06
Precision	92.78	64.20
F1	92.62	70.45

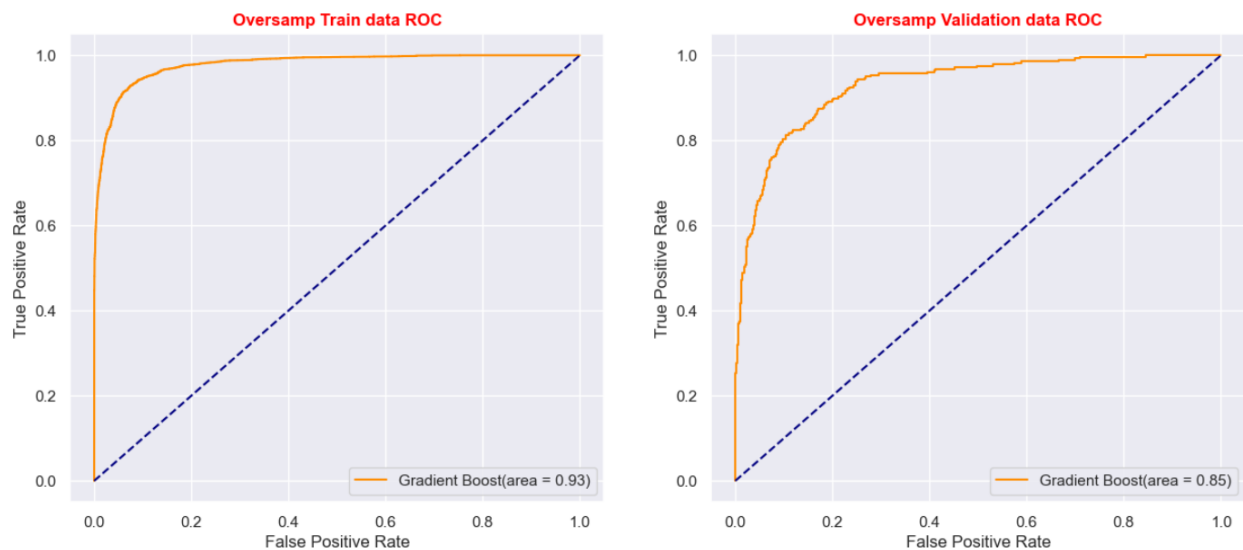


Fig 50: Gradient Boost AUC_ROC oversampled data

- The model delivers a training recall of 92.46%, indicating Low bias and strong pattern learning on known data.
- On the validation set, recall drops to 78.06%, which suggests high variance and poor generalization performance.
- However, the cross-validation recall average of 92%, same as the training score, reveals low variance across folds.
- A ROC AUC of 0.93 on the training set and 0.85 on the validation set indicates the model generalizes moderately across datasets but has scope of improvement.

10. XGBOOST:

	XGB_oversamp_Train	XGB_oversamp_Validation
Accuracy	99.96	97.03
Recall	99.94	88.13
Precision	99.98	93.87
F1	99.96	90.91

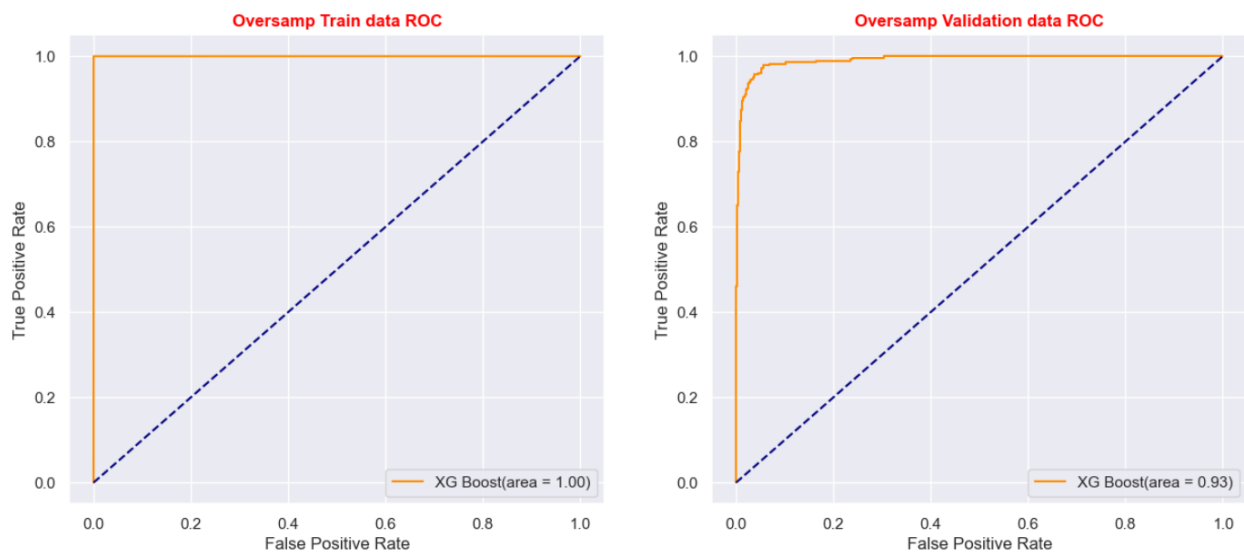


Fig 51: XGBoost AUC_ROC oversampled data

- The model delivers a training recall of 99.94%, indicating Low bias and strong pattern learning on known data.
- On the validation set, recall drops to 88.13%, which suggests moderate variance and moderate generalization performance.
- However, the cross-validation recall average of 98%, notably lower than the training score, reveals low variance across folds.
- A perfect ROC AUC of 1.00 on the training set suggests the model may be overfitting.

COMPAIRING RESULTS OF MODELS ON OVERSAMP DATA:

	Accuracy	Recall	Precision	F1
LR_oversamp_Train	80.45	80.12	80.65	80.38
NB_oversamp_Train	74.81	84.56	70.76	77.05
LDA_oversamp_Train	80.47	79.98	80.78	80.38
KNN_oversamp_Train	98.35	99.98	96.81	98.37
DT_oversamp_Train	100.00	100.00	100.00	100.00
Bag_oversamp_Train	99.94	99.90	99.97	99.94
RF_oversamp_Train	100.00	100.00	100.00	100.00
AB_oversamp_Train	87.61	87.47	87.71	87.59
GB_oversamp_Train	92.63	92.46	92.78	92.62
XGB_oversamp_Train	99.96	99.94	99.98	99.96

	Accuracy	Recall	Precision	F1
LR_oversamp_Validation	80.57	81.29	45.66	58.47
NB_oversamp_Validation	67.19	85.25	32.11	46.65
LDA_oversamp_Validation	80.39	81.29	45.38	58.25
KNN_oversamp_Validation	93.83	98.56	73.66	84.31
DT_oversamp_Validation	93.95	86.69	79.28	82.82
Bag_oversamp_Validation	95.70	87.41	87.10	87.25
RF_oversamp_Validation	97.88	92.45	94.83	93.62
AB_oversamp_Validation	84.26	74.46	52.27	61.42
GB_oversamp_Validation	88.98	78.06	64.20	70.45
XGB_oversamp_Validation	97.03	88.13	93.87	90.91

Balanced Performance:

Logistic Regression, Naïve Bayes, LDA shows balanced performance.

Overfitting:

All the other models show the characteristics of overfit in different ways.

Next steps:

XGBoost, Random Forest, and Bagging can be further tuned at a later stage to explore potential performance improvements.

MODEL TUNING:

RandomizedSearchCV has been used to fine-tune the model's hyperparameters

XGBOOST TUNED ON ACTUAL DATA:

HYPERPARAMETERS:

subsample: 0.6, reg_lambda: 0.5, reg_alpha: 0.5, n_estimators: 450, max_depth: 6, learning_rate: 0.05, gamma: 0.48000000000000004, colsample_bytree: 0.9, colsample_bynode: 1, colsample_bylevel: 1

	XGB_actual_train	XGB_actual_val
Accuracy	95.18	96.13
Recall	90.44	82.73
Precision	99.91	93.50
F1	94.94	87.79

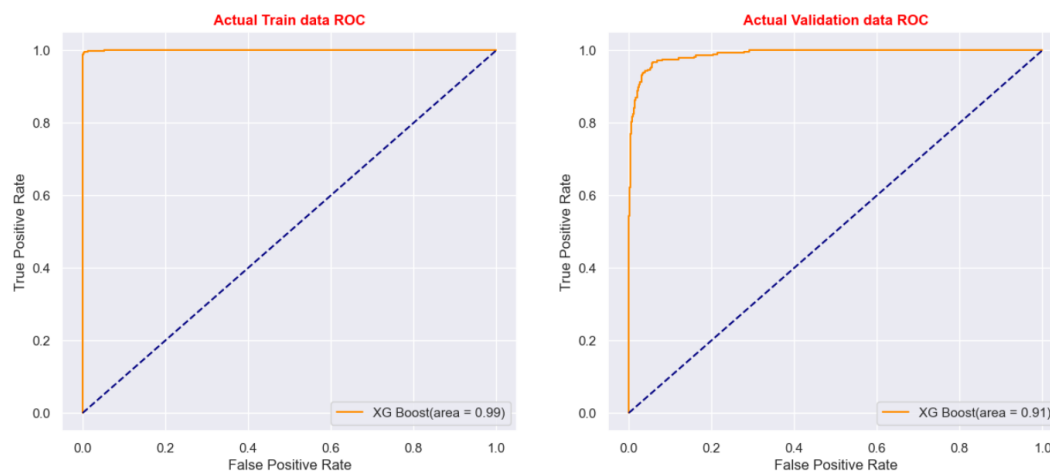


Fig 52: XGBoost AUC_ROC Tuned on actual data

- The model delivers a training recall of 90.44%, indicating Low bias and strong pattern learning on known data.
- On the validation set, recall drops to 82.13%, which suggests low variance and good generalization performance.
- Train ROC AUC of 0.99 and validation ROC of 0.91 indicate strong performance overall, but the near-perfect train score raises concerns about potential overfitting.

XGBOOST TUNED ON OVERSAMPLED DATA:

HYPERPARAMETERS:

'subsample': 0.7, 'reg_lambda': 0.7, 'reg_alpha': 0.6, 'n_estimators': 100, 'max_depth': 10,
'learning_rate': 0.1, 'gamma': 0.25, 'colsample_bytree': 0.8, 'colsample_bynode': 0.9,
'colsample_bylevel': 0.7

	XGB_oversamp_train	XGB_oversamp_val
Accuracy	99.88	96.43
Recall	99.81	88.85
Precision	99.95	89.82
F1	99.88	89.33

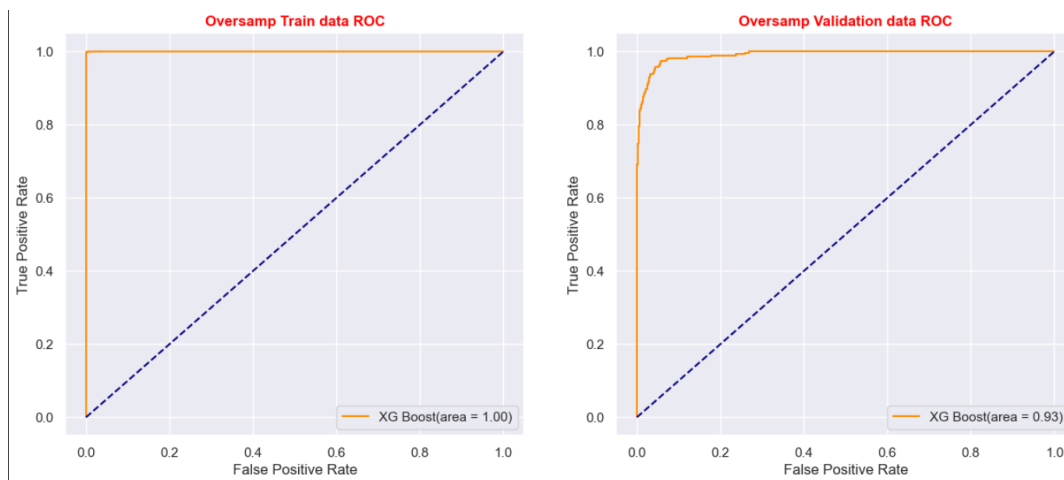


Fig 53: XGBoost AUC_ROC Tuned on oversampled data

- The model delivers a training recall of 99.81%, indicating Low bias and strong pattern learning on known data.
- On the validation set, recall drops to 88.85%, which suggests moderate variance and moderate generalization performance.
- A perfect ROC AUC of 1.00 on the training set suggests the model may be overfitting.

OBSERVATION:

- Both the XGBoost tuned models seems to be overfitting.

RANDOM FOREST TUNED ON NORMAL DATA:

Hyperparameters:

```
RandomForestClassifier  
  
RandomForestClassifier(class_weight='balanced', max_depth=9, max_features=0.9,  
                        max_samples=0.9, n_estimators=150, random_state=1)
```

	RF_actual_train	RF_actual_val
Accuracy	95.97	93.22
Recall	94.70	88.85
Precision	83.54	75.30
F1	88.77	81.52

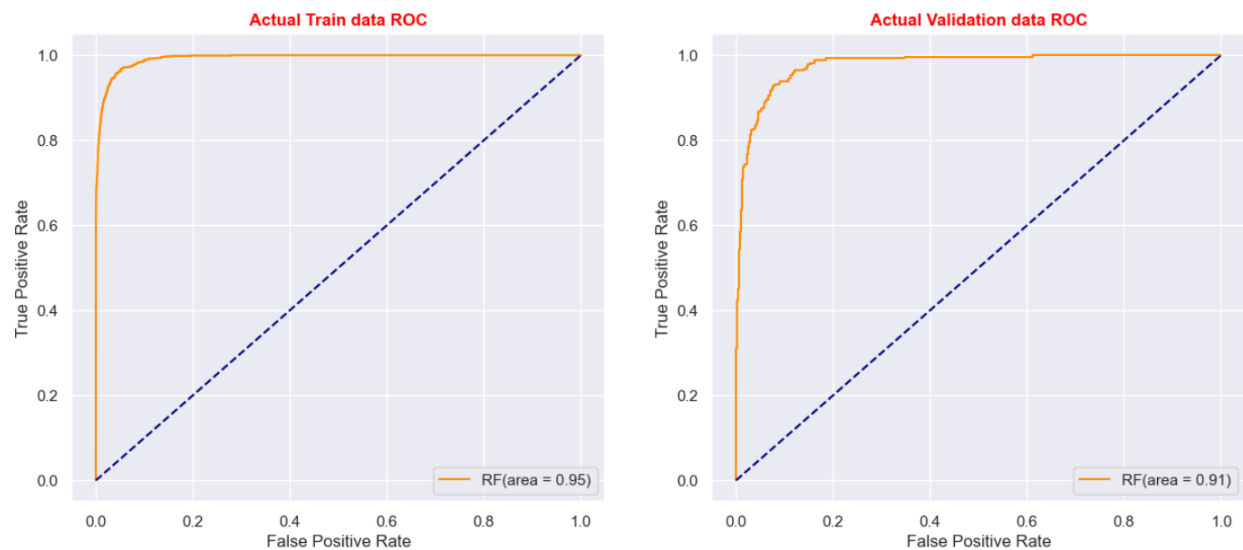


Fig 52: Random Forest AUC_ROC Tuned on actual data

- The model delivers a training recall of 94.70%, indicating Low bias and strong pattern learning on known data.
- On the validation set, recall drops to 88.85%, which suggests low variance and good generalization performance.
- A ROC AUC of 0.95 on the training set and 0.91 on the validation set suggests that this is a good model.

RANDOM FOREST TUNED ON OVERSAMPLED DATA:

Hyperparameters:

```
RandomForestClassifier  
RandomForestClassifier(max_depth=9, max_features=0.7, max_samples=0.7,  
                        n_estimators=125, random_state=1)
```

	RF_oversamp_train	RF_oversamp_val
Accuracy	96.52	92.62
Recall	96.50	87.41
Precision	96.55	73.64
F1	96.52	79.93

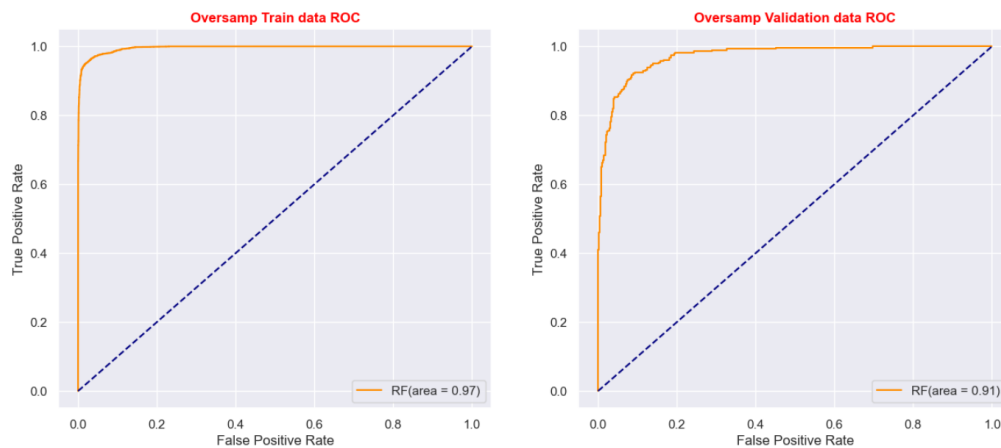


Fig 53: Random Forest AUC_ROC Tuned on oversampled data

- The model delivers a training recall of 96.50%, indicating Low bias and strong pattern learning on known data.
- On the validation set, recall drops to 87.41%, which suggests low variance and good generalization performance.
- A ROC AUC of 0.97 on the training set and 0.91 on the validation set suggests that this is a good model.

OBSERVATION:

Both tuned Random Forest models performed well, but the one trained on actual data showed better generalization.

BAGGING TUNED ON ACTUAL DATA:

Hyperparameters:

```
BaggingClassifier
BaggingClassifier(max_features=0.9, max_samples=0.8, n_estimators=60,
                  random_state=1)
```

	Bagging_actual_train	Bagging_actual_val
Accuracy	99.96	97.34
Recall	99.76	88.13
Precision	100.00	95.70
F1	99.88	91.76

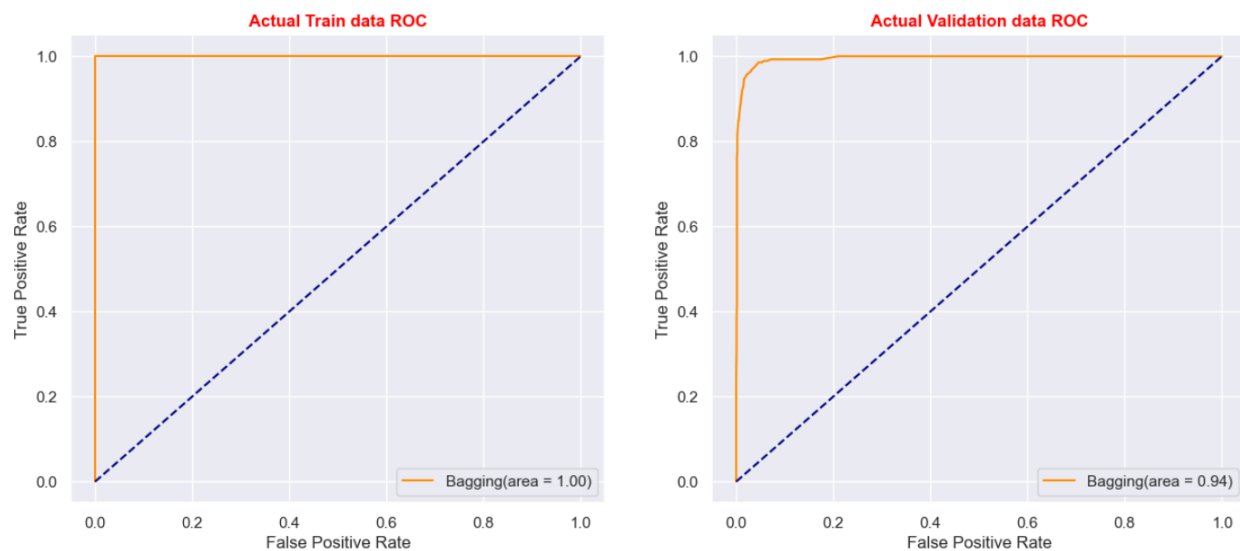


Fig 54: Bagging AUC_ROC Tuned on actual data

- The model delivers a training recall of 99.76%, indicating Low bias and strong pattern learning on known data.
- On the validation set, recall drops to 88.13%, which suggests moderate variance and moderate generalization performance.
- A perfect ROC AUC of 1.00 on the training set suggests the model may be overfitting.

BAGGING TUNED ON OVERSAMPLED DATA:

Hyperparameters:

```
BaggingClassifier
BaggingClassifier(estimator=DecisionTreeClassifier(random_state=1),
                  max_features=0.8, max_samples=0.6, n_estimators=140,
                  random_state=1)
```

	Bagging_oversamp_train	Bagging_oversamp_val
Accuracy	99.98	97.40
Recall	99.97	92.81
Precision	100.00	91.81
F1	99.98	92.31

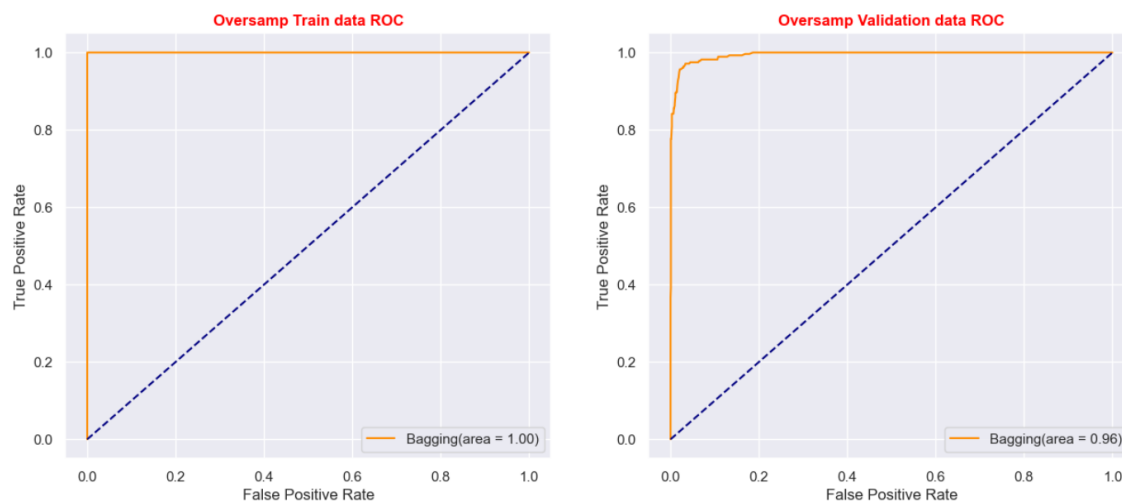


Fig 55: Bagging AUC_ROC Tuned on oversampled data

- The model delivers a training recall of 99.97%, indicating Low bias and strong pattern learning on known data.
- On the validation set, recall drops to 92.81%, which suggests low variance and good generalization performance.
- A perfect ROC AUC of 1.00 on the training set suggests the model may be overfitting.

OBSERVATION:

- Both the Bagging tuned models seems to be overfitting.

COMPARING TUNED MODELS:

	Accuracy	Recall	Precision	F1
XGB_actual_train	95.18	90.44	99.91	94.94
XGB_actual_val	96.13	82.73	93.50	87.79
XGB_oversamp_train	99.88	99.81	99.95	99.88
XGB_oversamp_val	96.43	88.85	89.82	89.33
RF_actual_train	95.97	94.70	83.54	88.77
RF_actual_val	93.22	88.85	75.30	81.52
RF_oversamp_train	96.52	96.50	96.55	96.52
RF_oversamp_val	92.62	87.41	73.64	79.93
Bagging_actual_train	99.96	99.76	100.00	99.88
Bagging_actual_val	97.34	88.13	95.70	91.76
Bagging_oversamp_train	99.98	99.97	100.00	99.98
Bagging_oversamp_val	97.40	92.81	91.81	92.31

- XGBoost and Bagging show overfitting on both actual and oversampled datasets.
- Random Forest performs well across both, ranking first and second, with the version trained on actual data delivering the best overall results.

BEST MODELS PERFORMANCE ON THE TEST DATA:

	RF_actual_train	RF_actual_val	RF_actual_test
Accuracy	95.97	93.22	93.02
Recall	94.70	88.85	89.38
Precision	83.54	75.30	74.39
F1	88.77	81.52	81.20

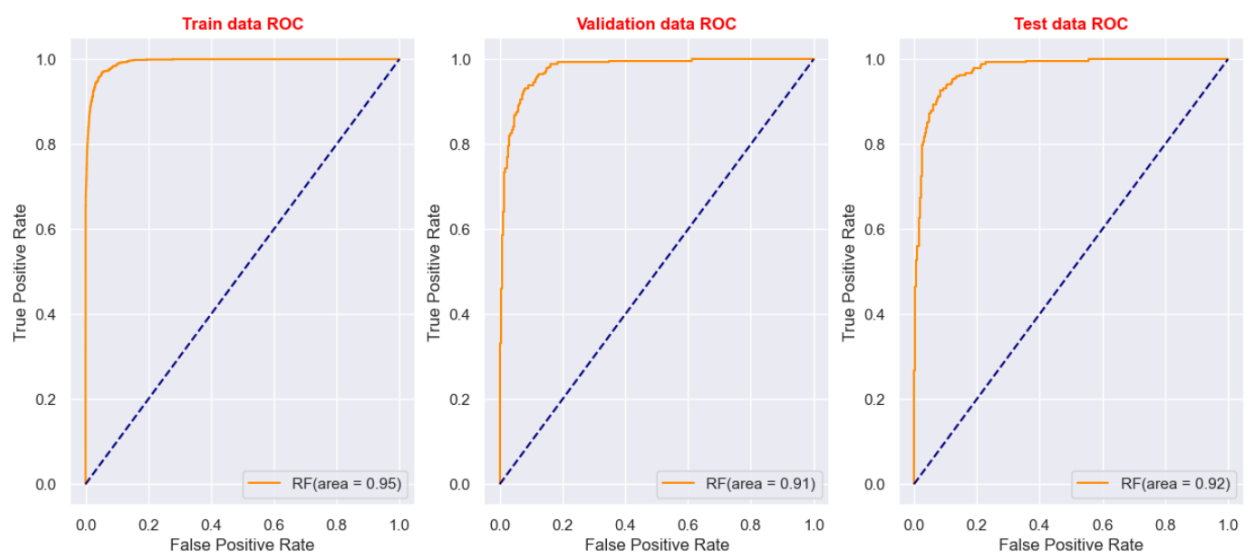


Fig 56: Random forest AUC_ROC Tuned on actual data

INTERPRETATION OF BEST MODEL:

- The Random Forest model built on actual data shows strong recall across train (94.70%), validation (88.85%), and test (89.38%) sets, indicating low bias and good generalization.
- Precision isn't optimal, but recall is our main focus for detecting churn.
- The ROC AUC scores—0.95 (train), 0.91 (validation), and 0.92 (test)—confirm this model is well-suited for production.

MODEL'S IMPLICATIONS ON BUSINESS:

- Capturing around 89% of churned customers on test data, this model enables the business to identify high-risk individuals early and take timely actions to retain them.
- By retaining these customers business can avoid revenue loss.
- Understanding the factors that pushes these customers toward churning helps business to fix this and improve its services towards customers which helps in long term.